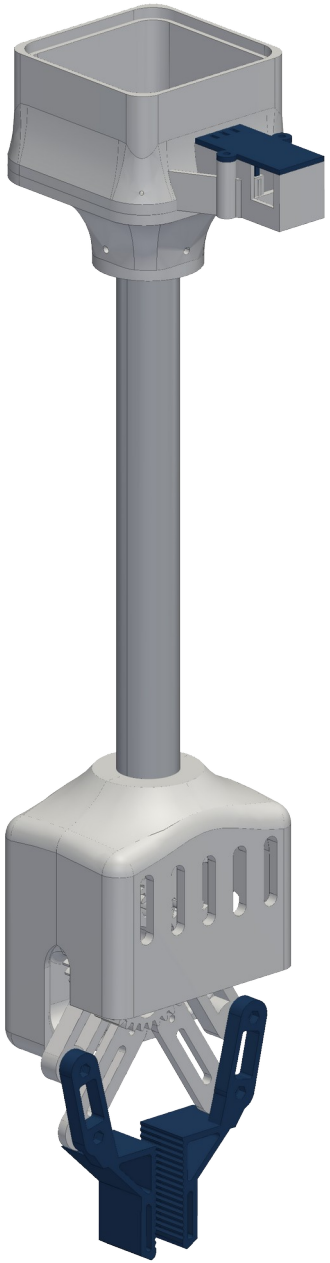
RoomCleaner

A four-cable ceiling robot that picks laundry off the floor

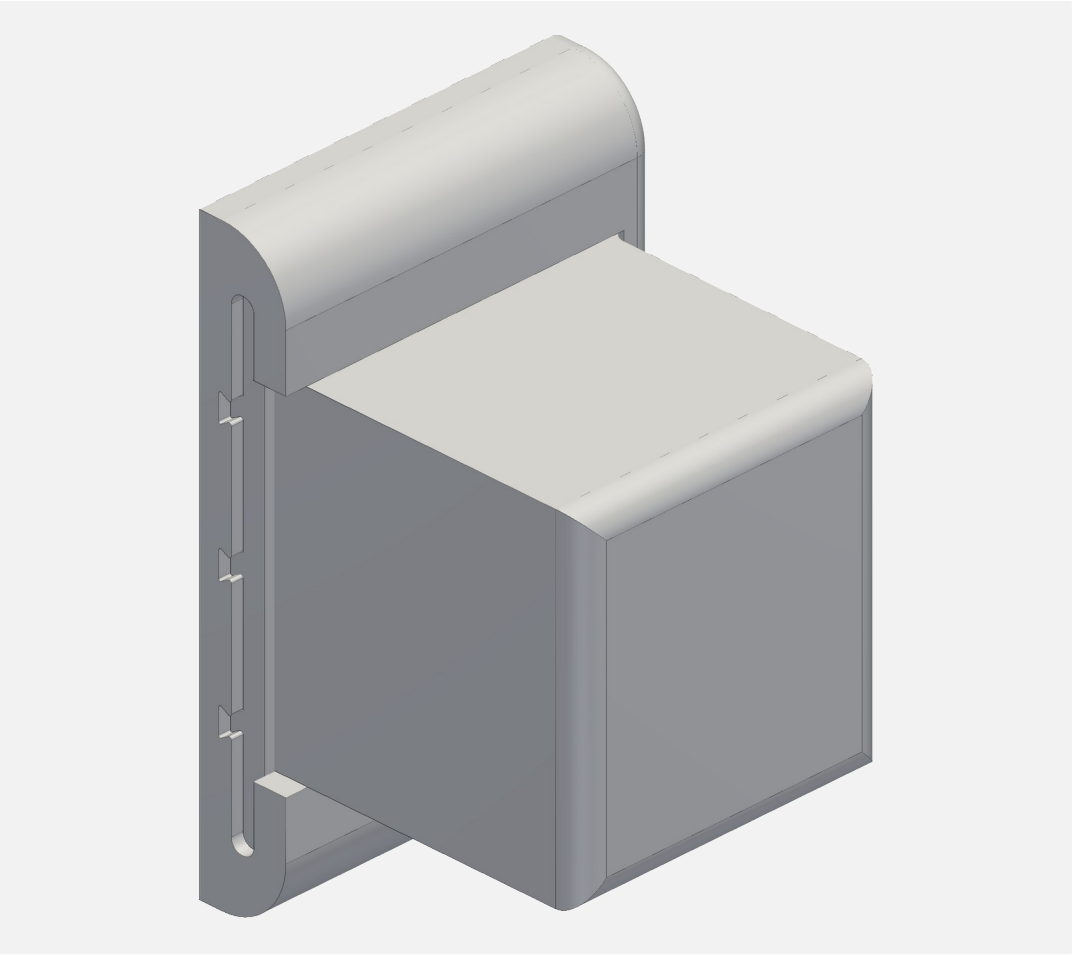
Jasper Buntinx
Technical presentation · October 27, 2026





Goal: pick laundry off the floor anywhere in an 11 × 9 ft room

- Pick clothes off the floor and drop them in a hamper, from anywhere in the room
- Four winches near the ceiling corners hold one claw on four cables
- Room 11 × 9 ft with a 9 ft ceiling
- Working load: a pair of jeans (0.9 kg) with 2 m/s² of acceleration; first test load 1 lb
- 3D-printed in PETG on a hobby budget, from parts I can buy and fix myself
- Safety: anchors in studs, one hard stop (the switched power strip), nobody under the robot



One of the four winch stations (CAD)

Outline

1. **How a cable robot works**
   Position vectors, statics, the winch
2. **From one box to modular stations**
   Wiring, a controller per winch, the slide-in housing, plug-in power, cooling and the horn-inspired fin
3. **Homing and calibration**
   Bead homing, why it limited the motion, the camera-based method
4. **The claw**
   Gripper evolution, the four-bar hands, overhead camera to claw camera, weight vs lift
5. **Manufacturing**
   Printed PETG parts, and a heat fin cut and drilled on a manual mill
6. **Electronics**
   Station board and a bring-up lesson
7. **Control logic**
   First logic vs final logic, one run end to end, the phone app, obstacles
8. **Status and next steps**
   First test, what is built, what I learned

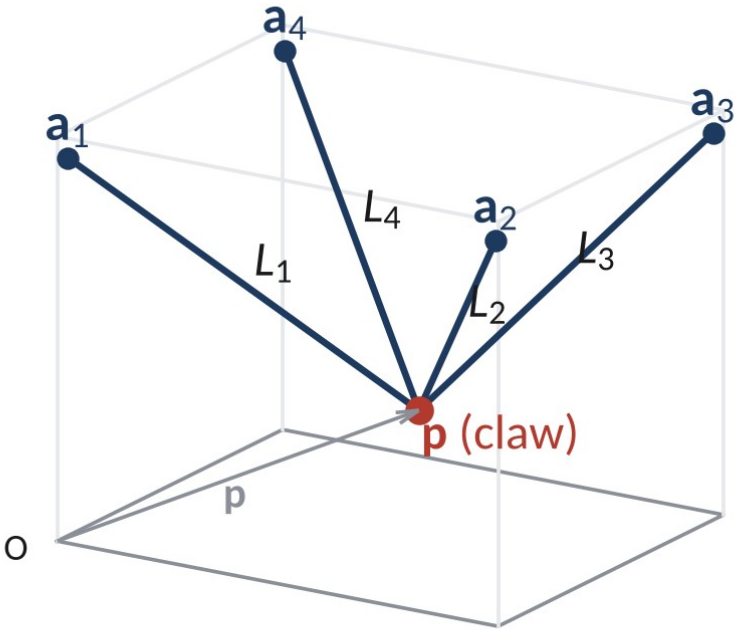
01

How a cable robot works

Four cables, one point, and the vectors that connect them

Each cable length is the distance from its anchor to the claw

- Room frame at a floor corner: anchor i (the pulley exit) at $\mathbf{a}_i$, the claw at $\mathbf{p}$
- Each cable is the vector from the claw to its anchor
- Its length is the distance between those two points
- To move, compute the four lengths at every point on the path
- Each spool turns by the change in length over its radius
- Closed form: no iteration is needed to command a move



Key equations

$$\mathbf{l}_i = \mathbf{a}_i - \mathbf{p}, \quad \mathbf{u}_i = \mathbf{l}_i / L_i$$

$$L_i = \|\mathbf{a}_i - \mathbf{p}\| = \sqrt{((a_{ix} - p_x)^2 + (a_{iy} - p_y)^2 + (a_{iz} - p_z)^2)}$$

$$\Delta\theta_i = \Delta L_i / r \quad (r = \text{spool radius with the wound line})$$

Cables only pull, so tension climbs as they flatten near the ceiling

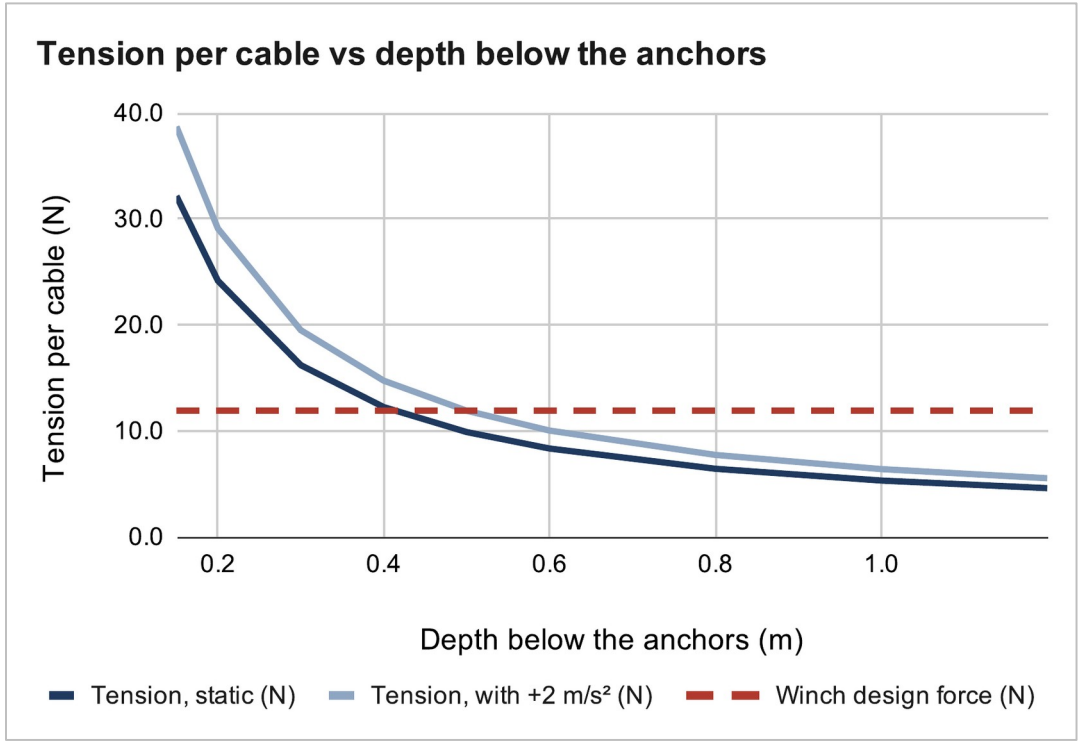
- Force balance: the cable pulls and the weight sum to zero
- Cables only pull: every tension must stay positive
- Centred claw, equal shares: tension rises as the cables flatten
- 32 N per cable 0.15 m below the anchors; 8.3 N at 0.6 m
- At 0.4 m: 12.2 N, right at the 11.9 N design force
- So the robot keeps 0.4 m below the anchors, 0.2 m from the walls

Key equations

$$\Sigma\, T_i\, \mathbf{u}_i + m(\mathbf{g} + \mathbf{a}) = \mathbf{0}, \quad T_i > 0$$

$$T = W / (4 \sin\varphi), \quad \sin\varphi = d / \sqrt{(d^2 + D_h{}^2)}$$

$$W = (0.45 + 0.454)\ \text{kg} \times 9.81\ \text{m/s}^2 = 8.87\ \text{N}$$



Source: calculation sheet (Google Sheets), “Cable tension” tab. Calculated, not measured.

At the driver's 1.06 A limit, each winch pulls 18 to 24 N

- NEMA 17 stepper, printed spool, outlet pulley
- Current limit set by the driver's reference voltage: 0.85 V → 1.06 A
- Running torque < holding torque: 0.7 assumed, the pull test measures it
- Smaller spool: more pull per amp, more turns per metre
- v3 spool: 24 N empty, 18 N full → 11.9 N design (÷ 1.5)

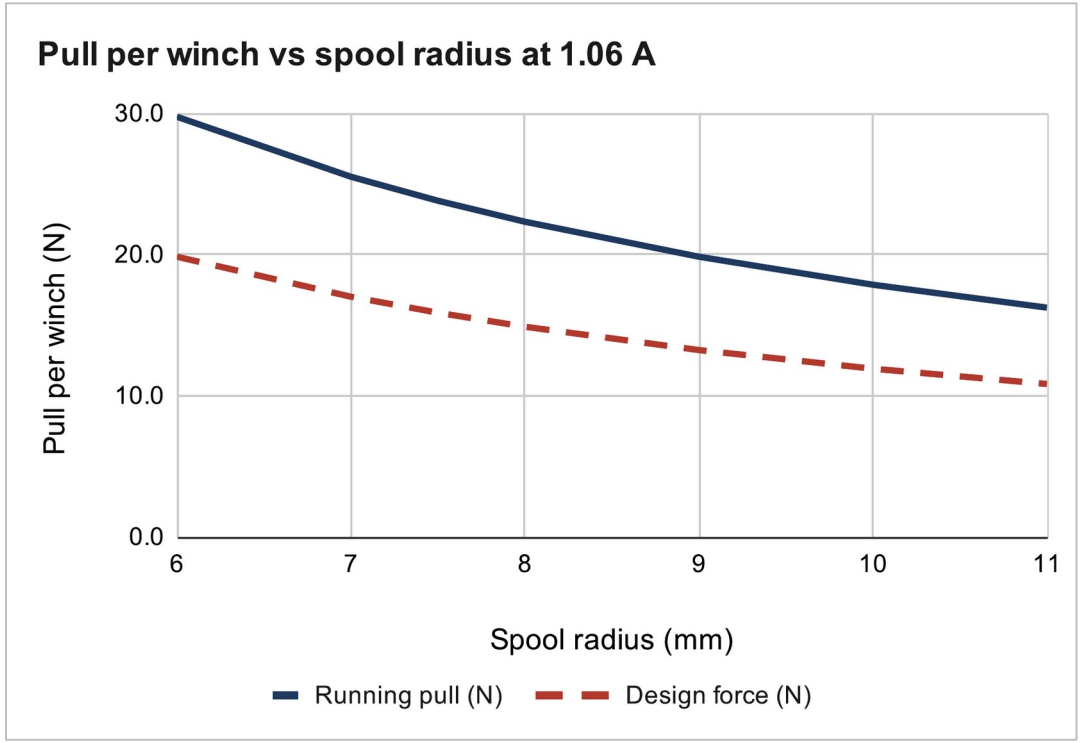
Key equations

$$I = V_{ref} / (8\ R_s) = 0.85\ V / (8 \times 0.1\ \Omega) = 1.06\ A$$

$$\tau_{run} = 0.7 \cdot \tau_{hold} \cdot I / I_{rated} = 0.7 \times 0.36 \times 1.06 / 1.5 = 0.18\ N{\cdot}m$$

$$T = \tau_{run} / r:\ \ 23.8\ N \text{ at } r = 7.5\ mm,\ \ 17.9\ N \text{ at } r = 10\ mm$$

Current limit: Allegro MicroSystems, A4988 datasheet



Source: calculation sheet (Google Sheets), “Winch pull” tab. Calculated, not measured.

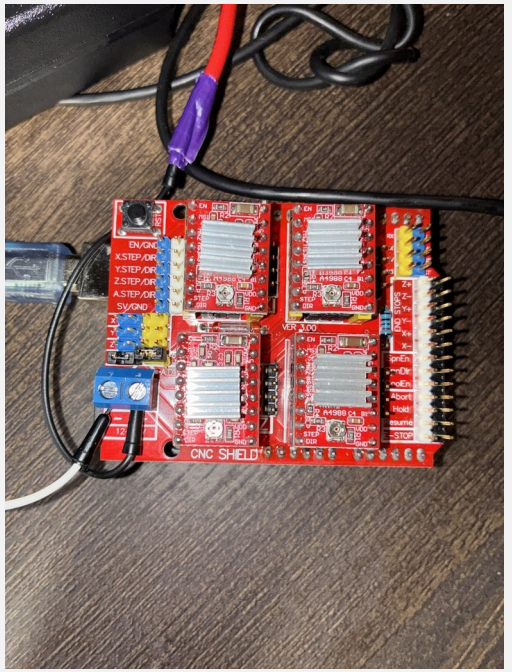
02

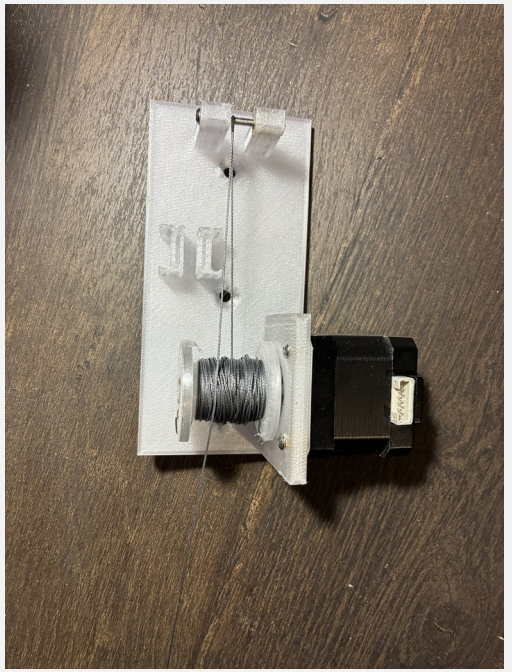
From one box to modular stations

How adding safety features changed the whole architecture

The first design drove all four motors from one central controller

- One Arduino Uno and a CNC shield with four A4988 drivers, in a single box
- The computer sends moves; the shield steps all four motors from one clock, so sync is free
- Each motor gets a long 4-wire cable from the box to its corner
- It worked on the bench: a gen-1 station lifted a test weight

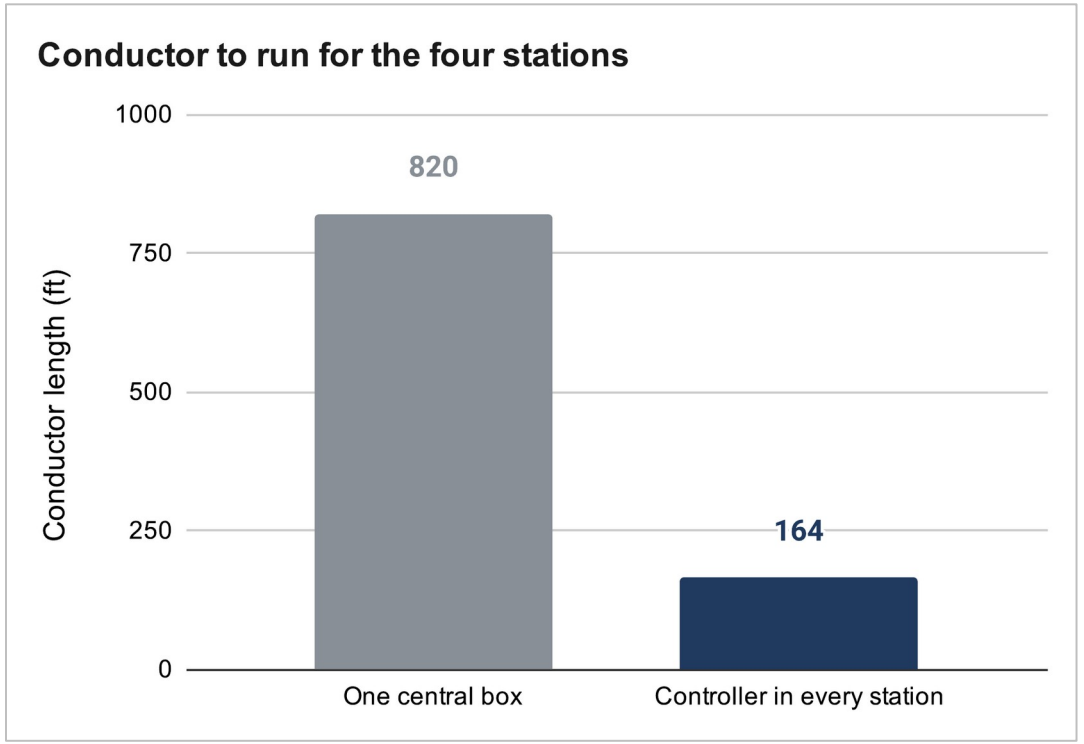




Gen 1: central shield with four drivers; spool plate on the wall

Sensors at every winch made central wiring five times longer

- I wanted more safety: a homing switch and an encoder at every winch, plus fault detection
- Per corner: motor 4 + switch 2 + encoder 4 = **10 conductors**, not 4
- Real cable routes are 6, 21, 23 and 32 ft (82 ft)
- The encoder's I²C bus is meant for centimetres, not 10 m runs
- Motor phase currents in long cables also radiate noise into the sensor lines



Source: calculation sheet (Google Sheets), “Wiring” tab. Calculated, not measured.

Key equations

$$\text{Conductor} = n \cdot \Sigma L: \quad 10 \times 82\ \text{ft} = 820\ \text{ft} \quad \text{vs} \quad 2 \times 82\ \text{ft} = 164\ \text{ft}$$

$$\Delta V = I \cdot R' \cdot 2L = 2\ \text{A} \times 4.016\ \text{m}\Omega/\text{ft} \times 64\ \text{ft} = 0.51\ \text{V}$$

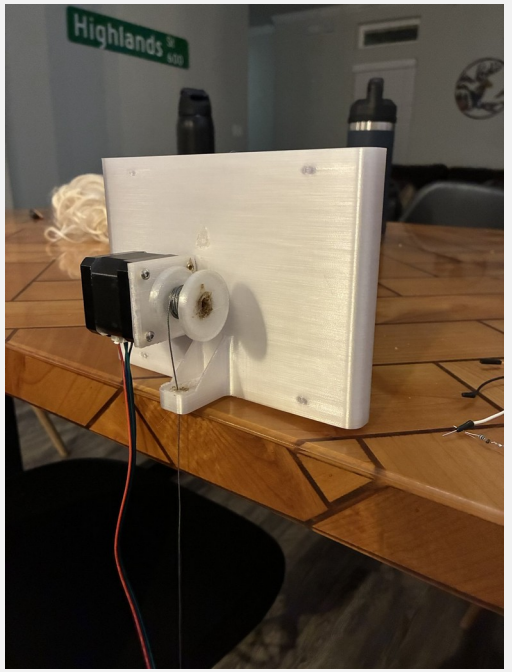
16 AWG on the longest run (32 ft); 2 A is illustrative

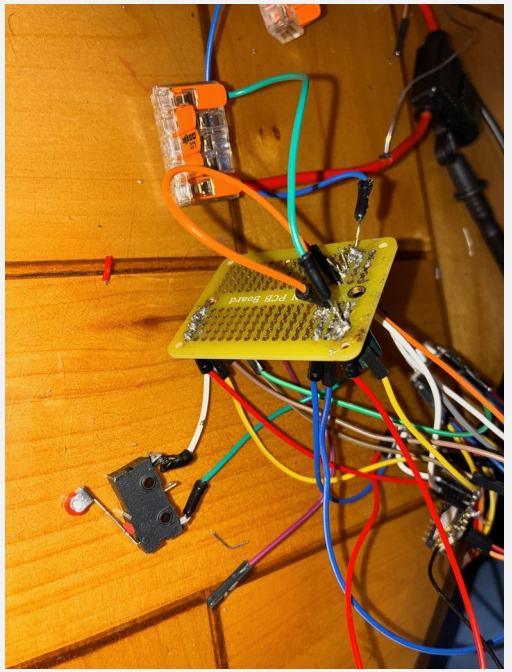
A controller in every winch cut the wiring to one power pair

- Each station: XIAO ESP32-C3 + A4988 driver + 5 V buck; only 12 V power travels, on 2 wires
- Switch and encoder wires stay inside the station, a few centimetres long
- The host computer plans the path and sends each station timed segments over Wi-Fi
- Each station runs its segments on its own clock; any lost link → hold position
- Cost: timing skew between stations becomes cable mismatch, so segments are time-stamped
- Benefit: about 5× less wire, and every station can be built and tested alone

Key equations

$$\Delta x = v \cdot \Delta t = 50\ \text{mm/s} \times 50\ \text{ms} = 2.5\ \text{mm of cable mismatch}$$

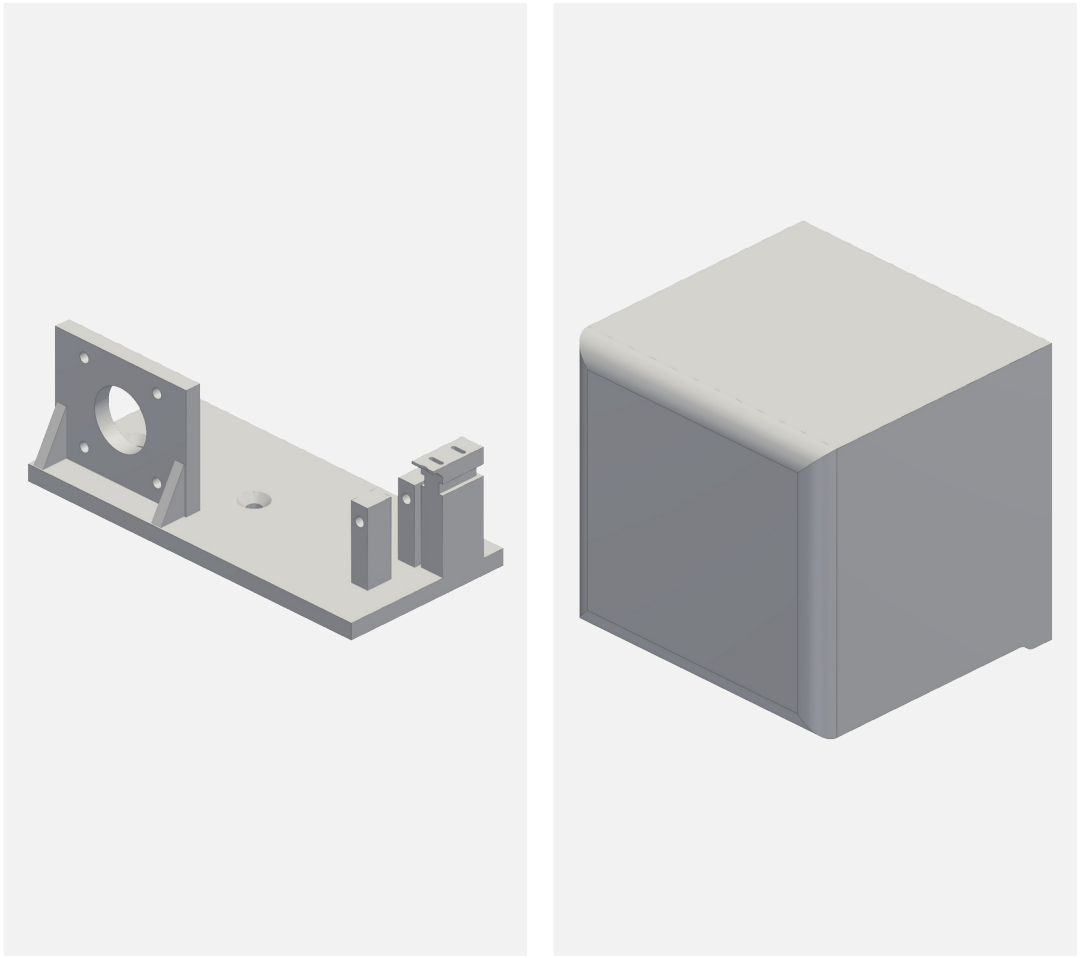




Gen 2: the station carries its own controller, driver and buck

Electronics at every corner turned the wall plate into a housing

- Gen 1 was a plate screwed into the wall: fine for a bare motor
- Each corner now carries a controller, driver, buck, switch and encoder
  - they need protection, a clean look and a path for heat
- So each station became a housing: motor, spool, pulley and electronics in one unit
- Same unit in all four corners: build one, test one, copy it



Gen-1 corner plate (left) vs the gen-2 station housing (right)

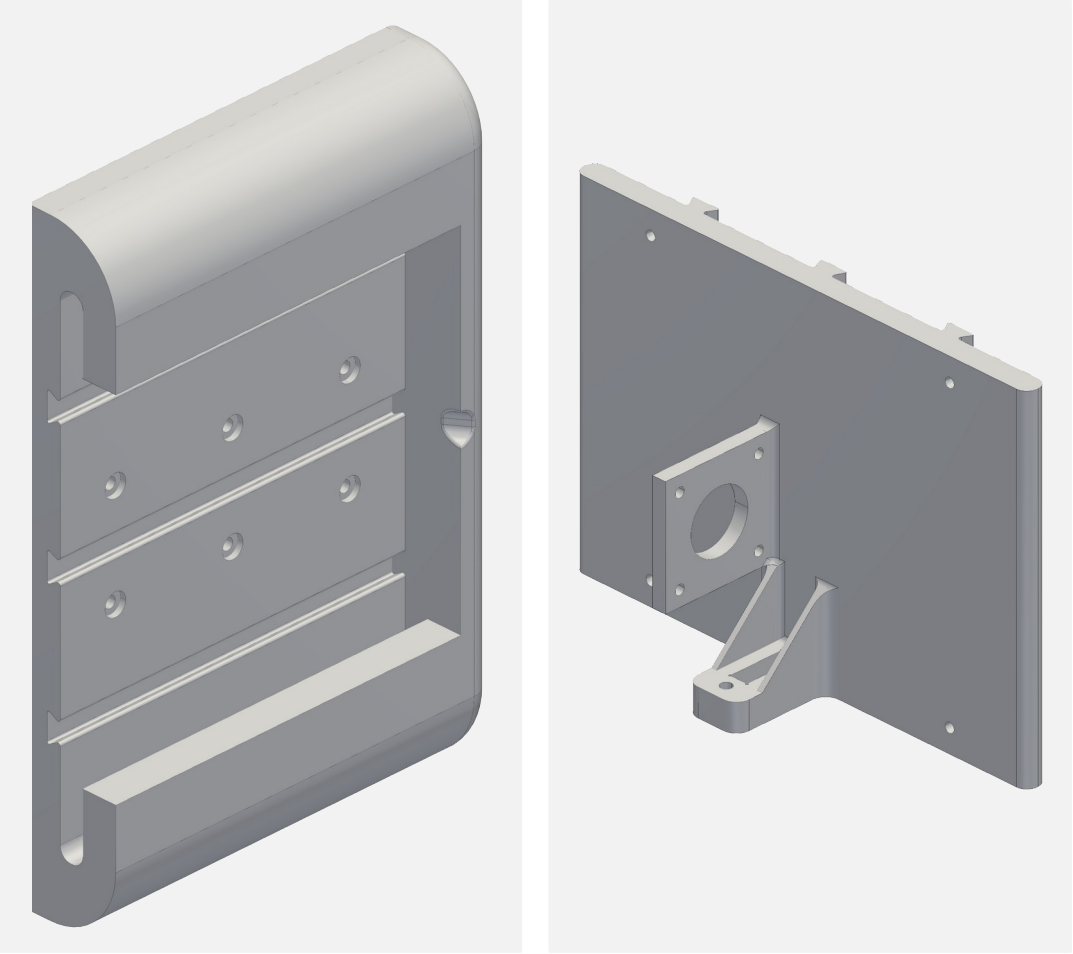
A slide-in mount lets a station come off without moving its anchor

- I take stations down often: to test, rework, repair or modify them
- Drilling in and out each time damages the wall and moves the anchor point
- A fixed wall frame goes into a stud once; the station slides in and locks
- Pull it out, rework it, slide it back: the frame keeps the anchor position
- Load: the line pulls the station away from the wall (pull-out and prying)
  - design target: SF 3 at 34 N; SolidWorks study planned

Key equations

$$SF = \sigma_{allow} / \sigma_{max} \geq 3 \rightarrow \sigma_{max} \leq 25\ MPa / 3 = 8.3\ MPa$$

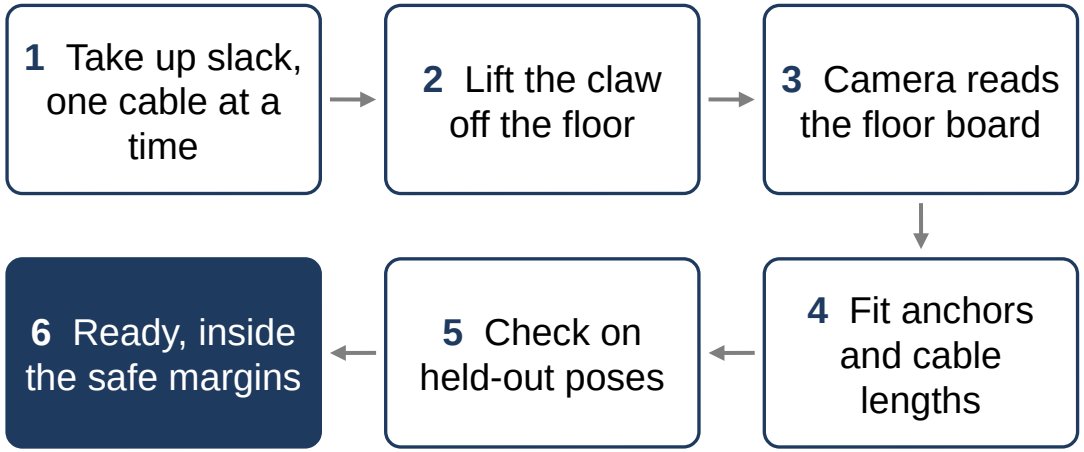
Allowable: derated PETG yield, 25 MPa

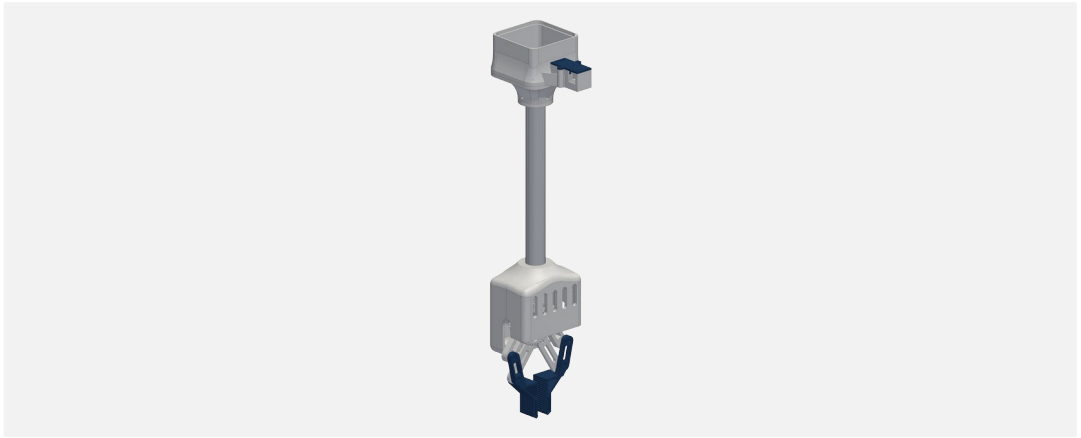


Fixed wall frame (left) and the slide-in piece (right)

The robot measures its own geometry every time it is set up

- Every time the stations go back up, the robot measures its own geometry
  - no tape measure, no numbers typed in, nothing for a user to get wrong
- The claw camera reads a printed floor board (a grid of tags) to get the claw's pose
- The robot moves through a set of poses and fits the anchor positions and cable lengths by least squares
- Works in any room the cables can reach, so it can become a consumer product

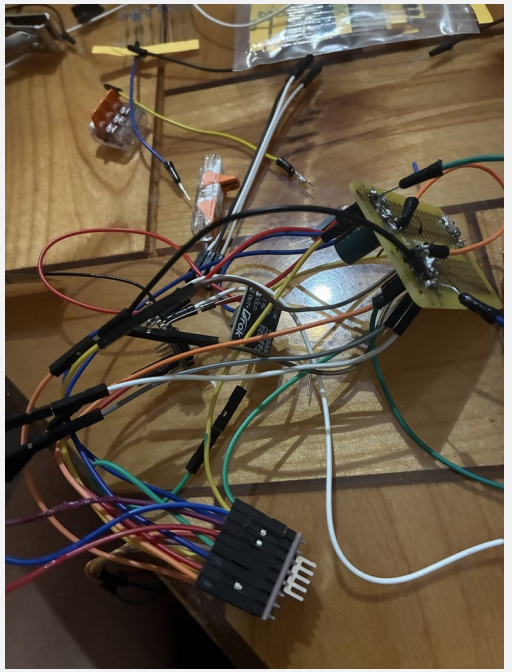


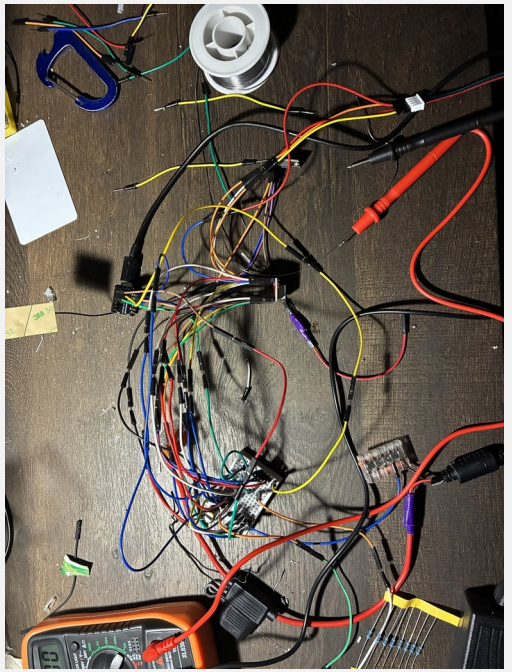


Startup calibration; the claw camera looks straight down at the board

Each station needs only one 12 V cable, so it unplugs cleanly

- Each station has one cable: a 16 AWG pair from a fused 12 V supply
- Plug in to use; unplug and store the cables when the robot is not needed
- Stations slide off the wall frames if wanted
- Runs cut at 9, 24, 26 and 35 ft (94 ft of a 100 ft roll), a fuse per branch at the supply
- The motor stays wired inside the station, so it is never unplugged under power
- The switched power strip is the one hard stop

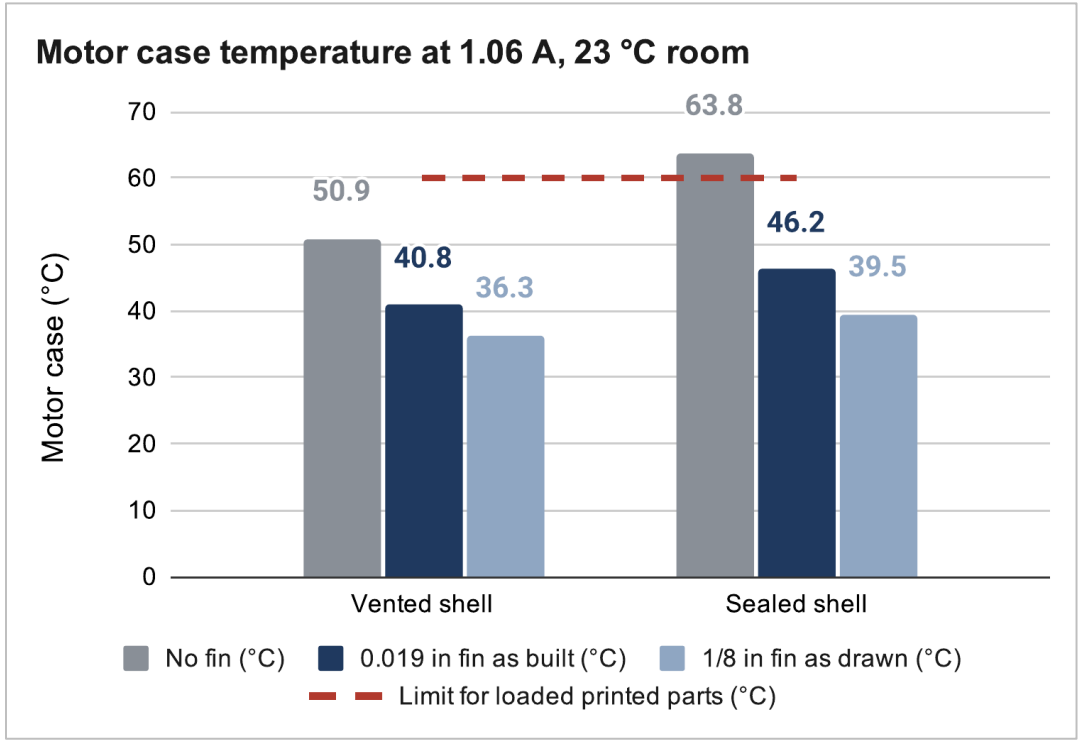




Inside a station: the buck converter and its wiring; only 12 V comes in

Without the fin, a sealed housing would push the motor past 60 °C

- A sealed printed housing traps the motor's heat
  - PETG at the loaded motor mount softens and creeps: 60 °C limit
- Fix: vents, plus an aluminium fin straight up through the shell
- Model: motor → shell air → room, the fin a second path out
- At 1.06 A: sealed 64 °C without the fin, 46 °C with it
- A 90-min hold test will measure it



Source: calculation sheet (Google Sheets), “Motor temperature” tab. Calculated, not measured.

Key equations

$$P_m = I^2 R = (1.06\ \text{A})^2 \times 1.8\ \Omega = 2.0\ \text{W}$$

$$\Delta T = q \cdot R_{th} \quad \text{(heat flow} \times \text{thermal resistance)}$$

$$\text{Fin in parallel with the shell: } 1/R = 1/R_{shell} + 1/R_{fin}$$

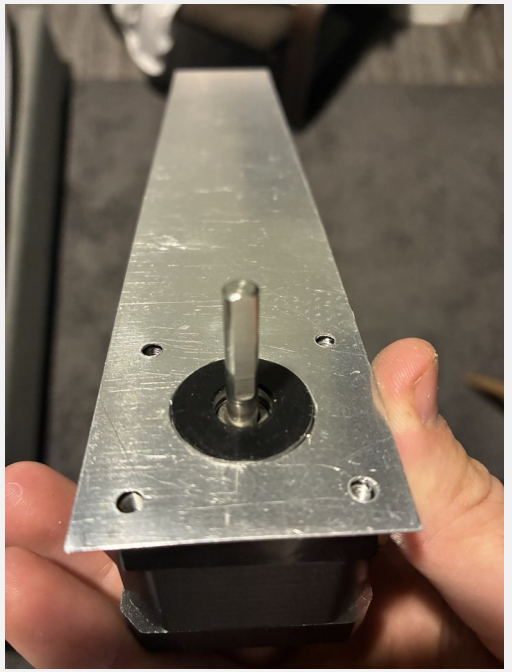
The fin copies an antelope horn: a thin blade that sheds heat

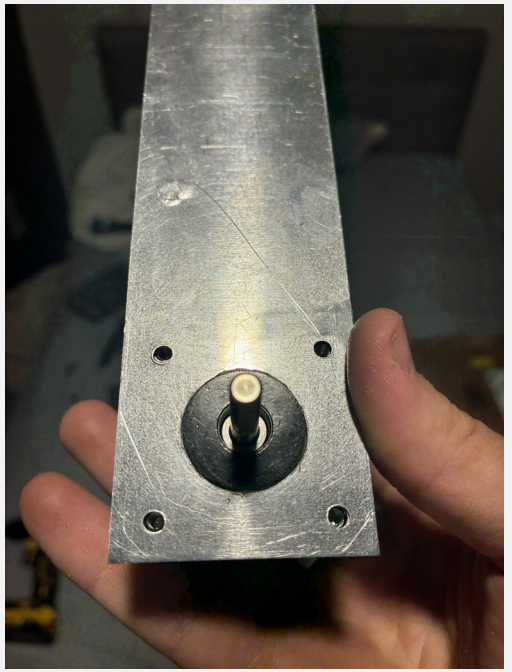
- Idea from my bio-inspired design class: antelope horns
- A horn is a heat exchanger: warm blood flows into its core, heat leaves through the surface
- The fin copies that: one tall, thin blade rising straight up from the heat source
  - heat conducts up from the motor and leaves along the length; the tip runs cooler, like a horn's
- Vertical, so warm air rises freely off both faces
- Motor width (44 mm): adds height, not width, to the station

Key equations

Horn and fin both shed heat by convection: $q = h \cdot A \cdot (T_s - T_\infty)$

Tall and thin: large area for little mass, $A = 2\,w\,L$





The fin on the motor: one tall blade rising from the heat source

Past 136 mm, the thin fin adds less than 5 % more cooling

- The far end runs cooler, so each extra mm does less
- **0.019 in sheet (built):** 95 % of a long fin's heat by **136 mm**
- **1/8 in plate (drawn):** 95 % by **349 mm**
- Built 200 mm above the cover: past 136 mm the sheet adds < 5 %
- Looks: horn-like, motor width (44 mm), matte black to radiate more

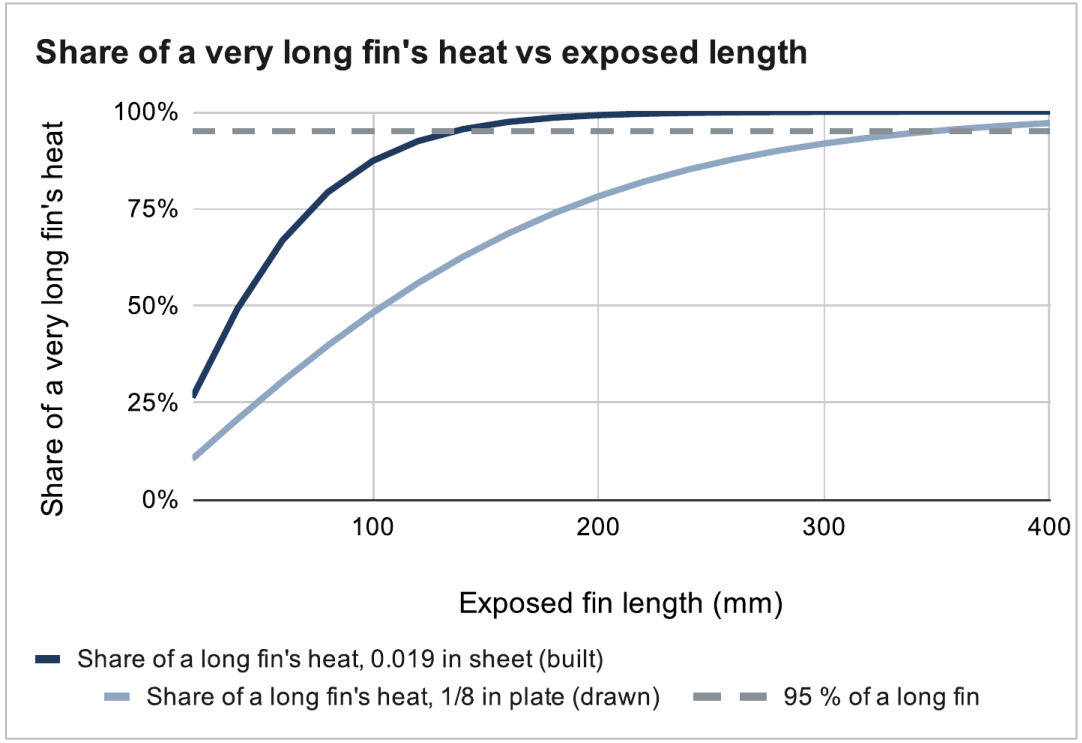
Key equations

$m = \sqrt{(2h / (k\,t))}, \quad \eta = \tanh(mL) / (mL)$

95 % length: $\tanh(mL) = 0.95 \rightarrow L = 1.83 / m$

$k = 160\ W/m{\cdot}K,\ h = 7\ W/m^2{\cdot}K,\ t = 0.48\ mm \rightarrow 136\ mm$

Fin equations: Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3



Source: calculation sheet (Google Sheets), “Heat fin” tab. Calculated, not measured.

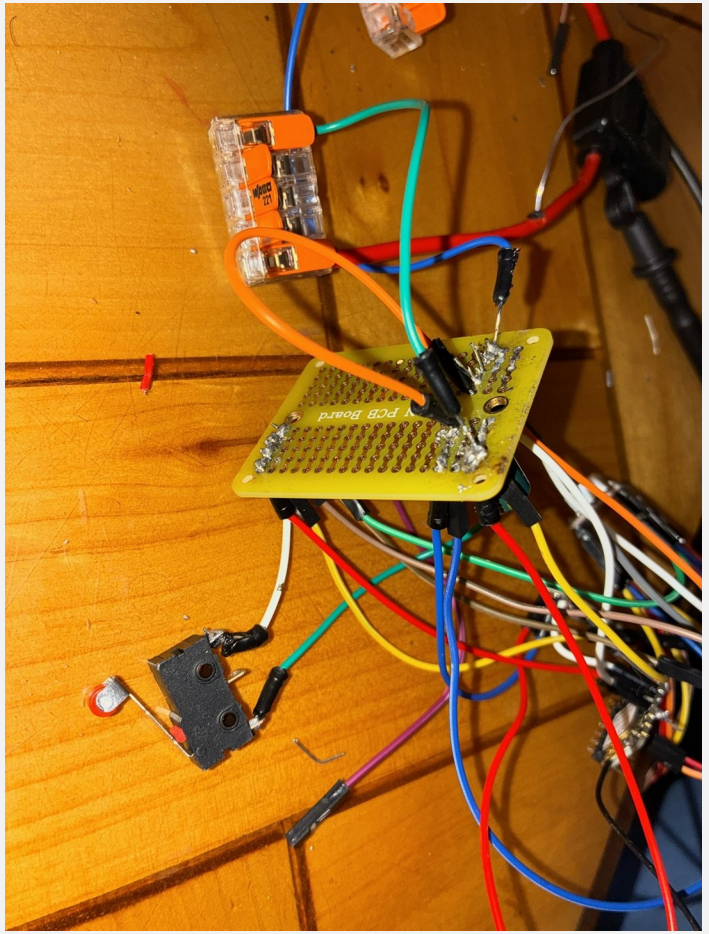
03

Homing and calibration

Why the first homing plan limited the robot, and what replaced it

The first plan homed each cable with a bead and a switch

- A bead on each cable trips a switch at the outlet → a known cable length
- The robot zeroes itself from that: set it up anywhere
- I expected that to be enough, possibly even without encoders
- Detail fixed early: the switch moved below the pulley, so the bead never rides over the pulley groove



KW12-3 homing switch on the bench

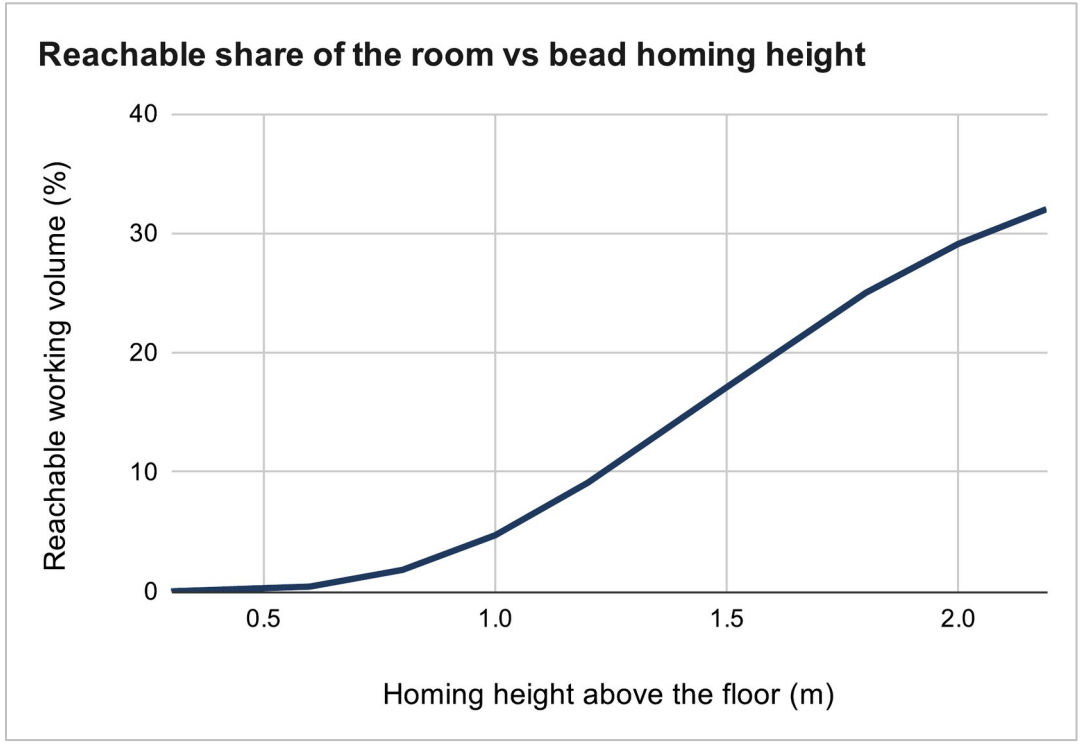
Bead homing at one point leaves under a third of the room reachable

- A bead cannot pass the outlet, so a cable can never be shorter than its homing length
- To home all four at one point, each bead sits at that point's length
  - then the claw can never get closer to any anchor than from that point
- Homing at the room centre 1 m up → about 5 % of the working volume reachable; 2 m up → about 29 %
- And vision alone knows cable length to ~20–25 mm, while a bead allows ~12 mm of overtravel: a bead search was not safe either

Key equations

$$L_i(\mathbf{p}) \geq L_{i,\text{home}} = \|\mathbf{a}_i - \mathbf{p}_{\text{home}}\| \quad \text{for every cable } i$$

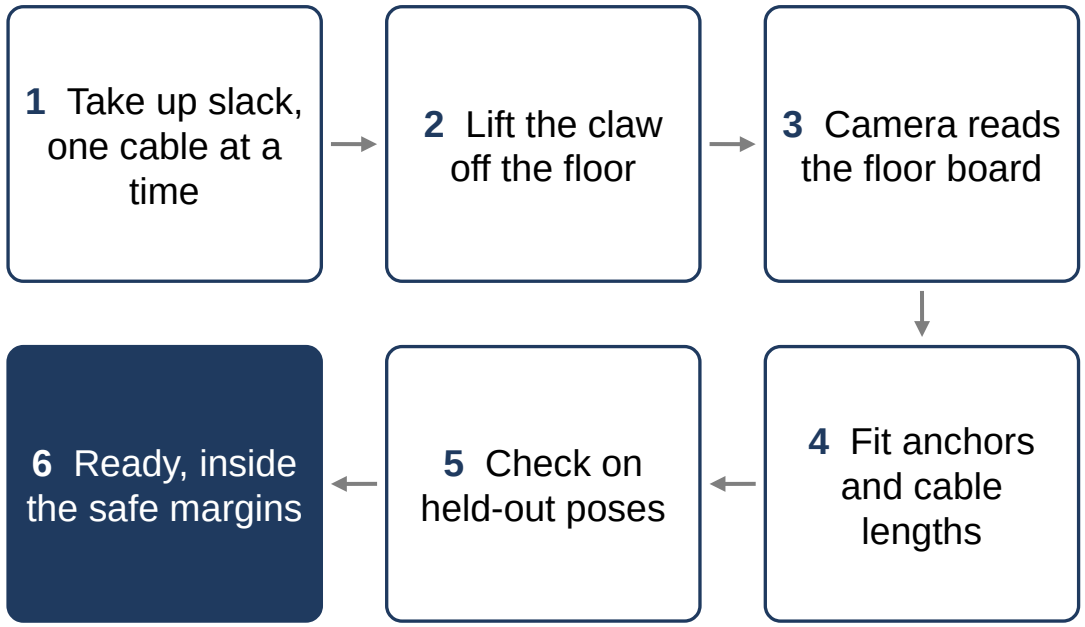
Reachable: every cable at or above its homing length



Source: calculation sheet (Google Sheets), “Bead homing” tab. Calculated, not measured.

Camera poses fit the geometry, and safety margins cover the error

- Startup: take up slack one cable at a time, lift, then look at the floor board from several poses
- Least-squares fit of the anchor positions and cable zero lengths; held-out poses check the fit
- Safety comes from margins, not precision
  - 0.4 m below the anchors and 0.2 m inside the walls unless the error bars allow more
- Beads stay as end-of-travel safety switches; encoders watch for lost steps



If the fit is not good enough: more poses, then fit again

Key equations

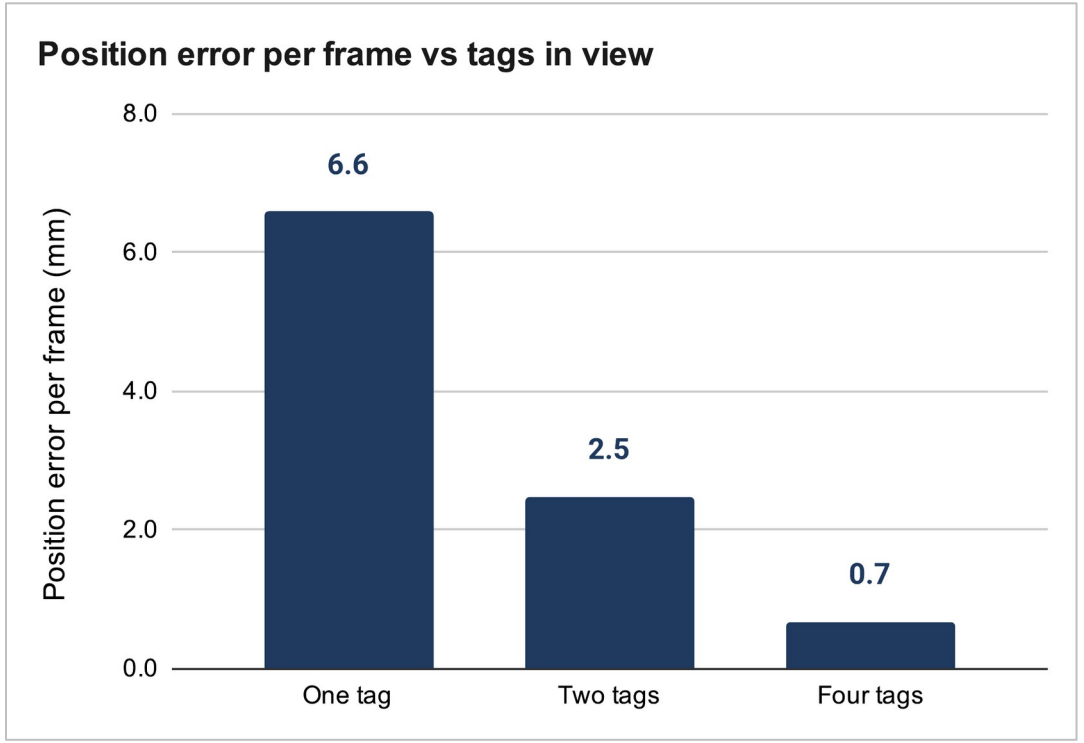
$$\min \text{ over } \mathbf{a}_i, L_{i0}: \quad \Sigma_k \Sigma_i \left(L_{i0} + d_{ik} - \|\mathbf{a}_i - \mathbf{p}_k\| \right)^2$$

$$\text{Full workspace only if } k \cdot \sigma \le \text{margin}$$

d: cable paid out at pose k (encoder); p: claw position from the camera

Seeing four tags instead of one cuts position error from 6.6 to 0.7 mm

- Angle error becomes position error through the lever arm
- One tag: 12.9 mrad × 0.51 m ≈ 6.6 mm per frame
- Four tags in view: about 0.7 mm
- The board is a printed grid at 250 mm pitch: cheap, and it works in any room
- Averaging frames helps only with random noise, not with a fixed bias



Source: calculation sheet (Google Sheets), “Floor board” tab. Calculated, not measured.

Key equations

$$\delta x = \sigma_\theta \cdot L = 12.9\ \text{mrad} \times 0.51\ \text{m} = 6.6\ \text{mm}$$

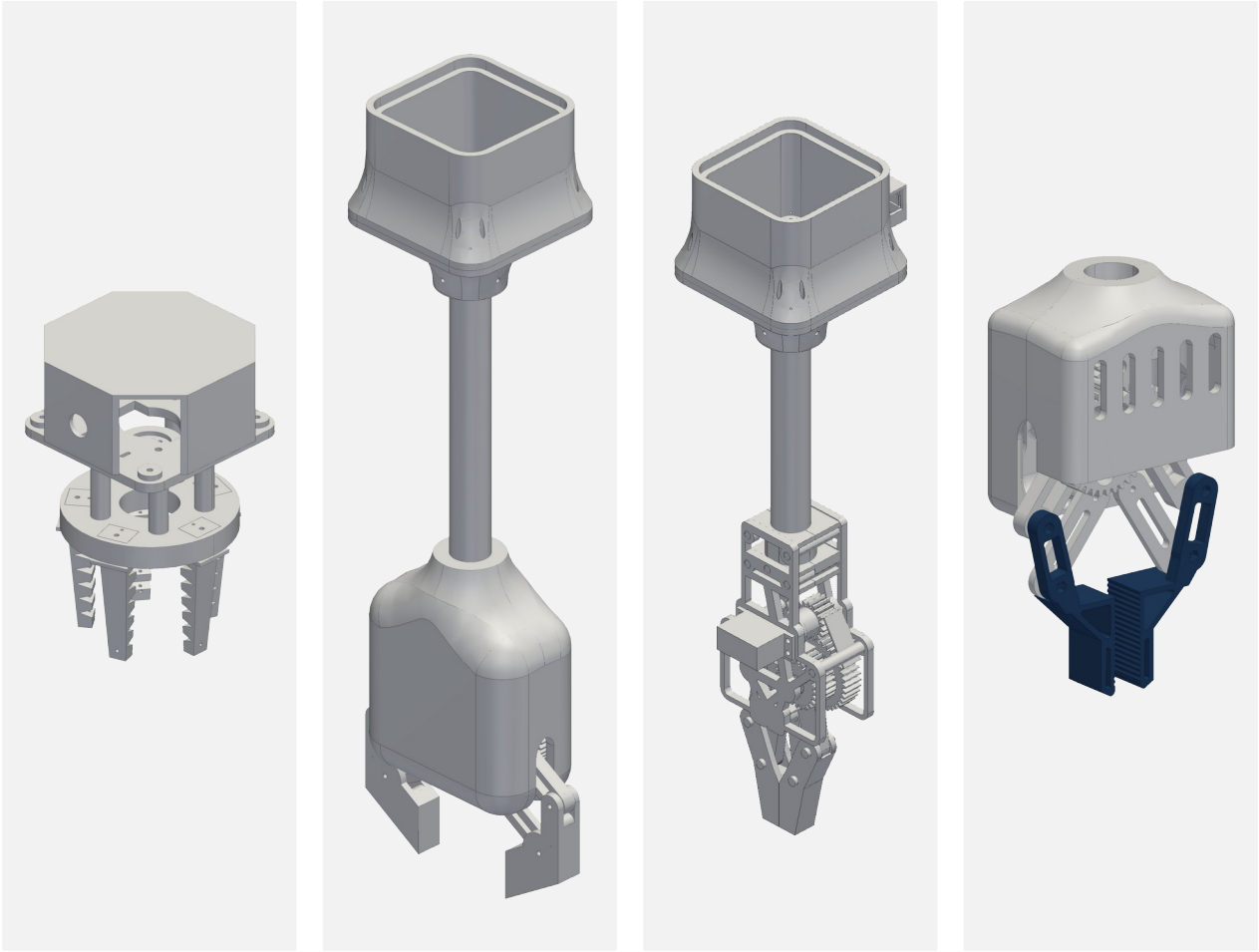
04

The claw

From fingers to a parallel-jaw gripper

Each claw generation fixed a problem found in the one before

- Five tendon-driven TPU fingers → a geared clamp → a lighter open frame → the final parallel-jaw gripper
- Each step fixed a problem I found:
  - the first clamp's jaws could not pinch thin fabric
  - the v6 clamp was about 4.6× its mass budget
  - v7 cut the gearbox frame to 143 g
- Final: my gear housing and arms, with hands designed to fit them



Fingers, v6 clamp, v7 open frame, final gripper

A geared four-bar keeps the jaws parallel through a 53 mm stroke

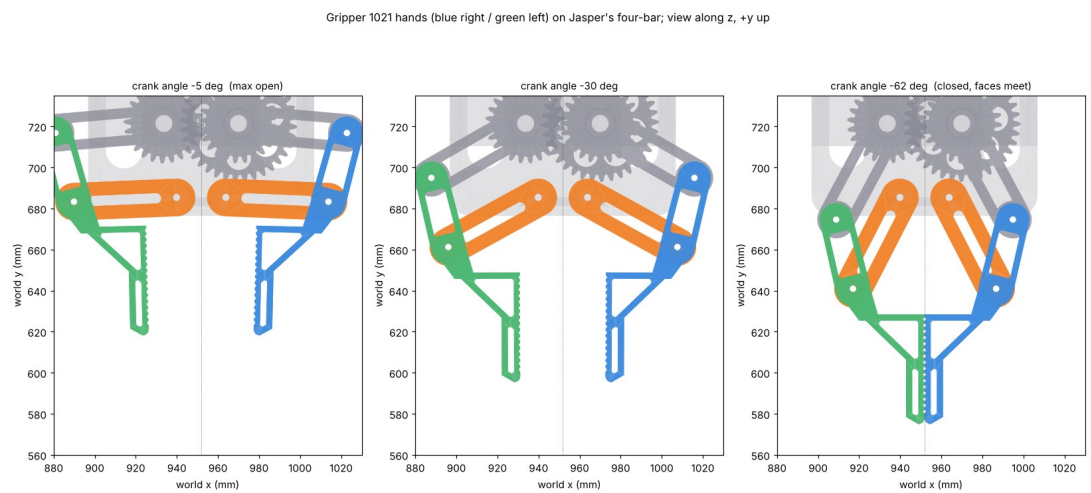
- Servo → 18T pinion → 36T/24T compound → 24T sync gear: mirror-image jaws
- Each hand: gear crank (53 mm) + swing arm (50 mm), a near-parallelogram four-bar
  - jaws stay parallel within about 1° over the stroke
- 53 mm opening; closing drops the tips ~38 mm: a scoop
- Gear teeth at servo stall: safety factor about 2
- Swept in CAD: no collisions from open to closed

Key equations

Gear ratio $i = 36 / 18 = 2$ (servo → crank)

Grip: $2 \mu N \geq m g \rightarrow N \geq 7.4$ N per jaw (1 lb, $\mu = 0.3$)

Lewis: $\sigma = F_t / (b m Y)$, $F_t = 2\tau / d$



Hands on the linkage: open, mid, closed

The first setup found laundry with one camera on the ceiling

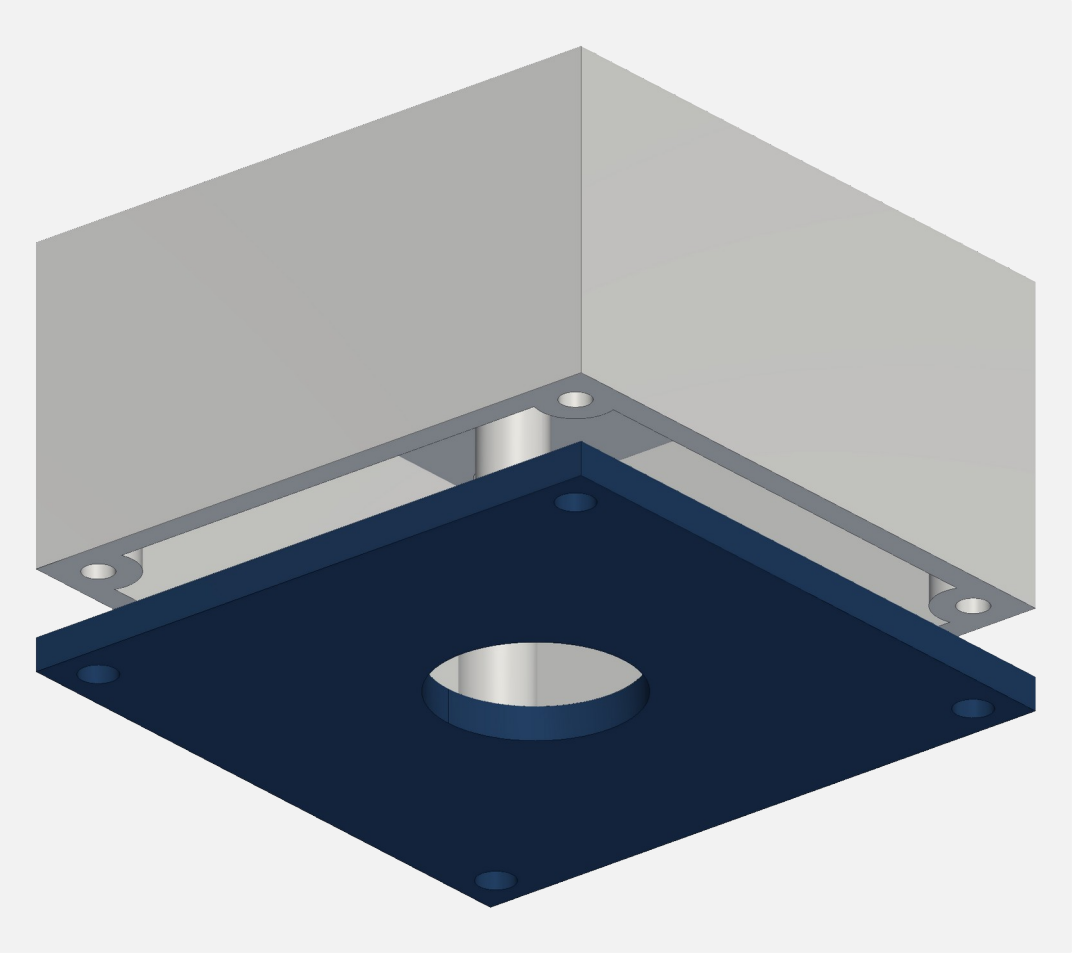
- One camera on the ceiling, looking straight down and framing the whole floor
- A detector that works from words (YOLO-World) finds “sock”, “towel”, “t-shirt” with no training
- Each item's bottom-centre pixel, where it meets the floor, becomes a floor position
  - straight-down camera: scale by the room size; tilted camera: a homography from four taped floor points
- The claw then flies to that point and grabs
- The camera board sat in a printed ceiling case: shell plus a screw-on lens bezel

Key equations

Straight down: $x = u \cdot W / w_{img}, \quad y = v \cdot D / h_{img}$

Tilted: $(x, y, 1) \propto \mathbf{H} \cdot (u, v, 1), \quad \mathbf{H}$ fitted from 4 floor points

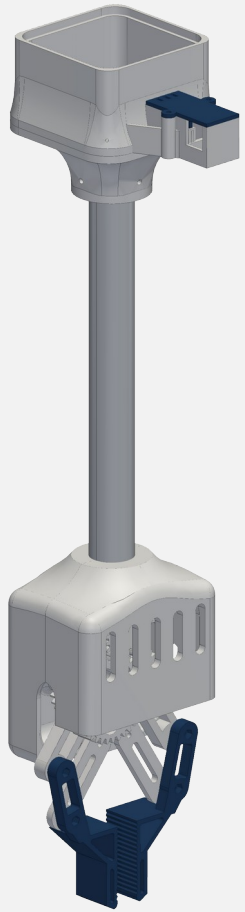
u, v: pixel; W × D: floor size; w × h: image size in pixels



Gen-1 ceiling camera case: shell and lens bezel (45 g PETG)

The camera moved from the ceiling to the claw, right at the grasp

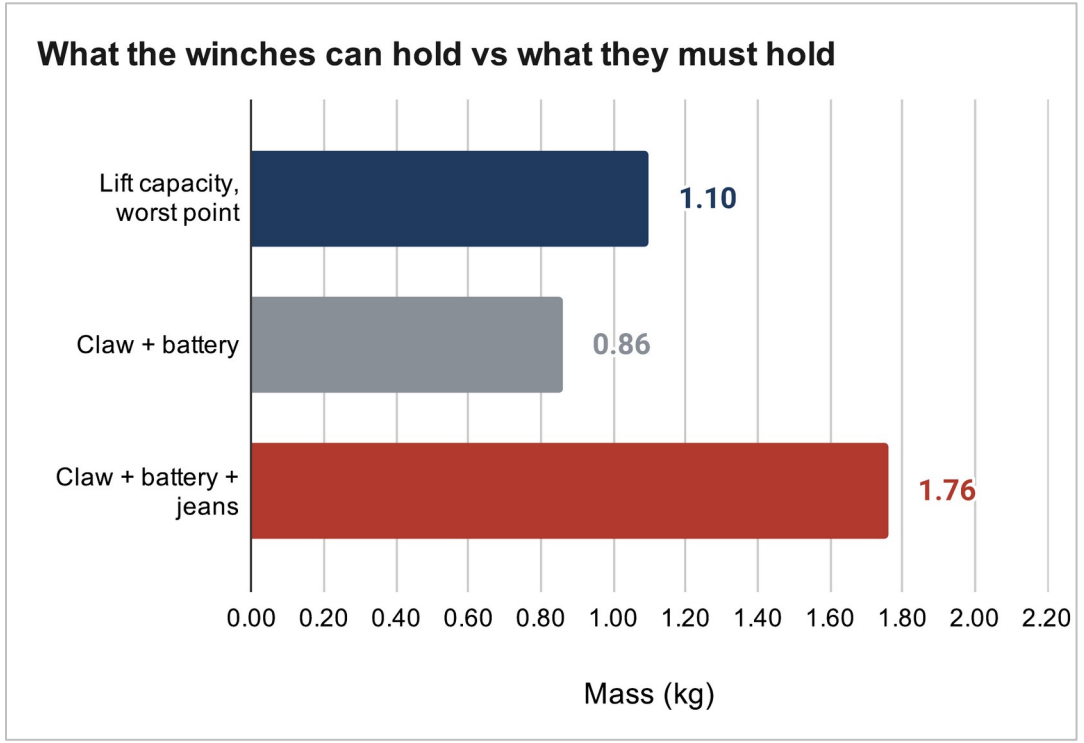
- The overhead camera was one more device to mount, wire and calibrate in every room
- The claw and cables hide the spot right where it grasps: it cannot see the grip
- Its floor map assumes a flat floor, but a pile of clothes has height
- Now: a XIAO ESP32-S3 Sense camera in a bay in the claw's housing, looking straight down the tube
- One camera, two jobs: it guides the grasp up close and reads the floor board for calibration
- Cost: it sees less of the room at once, so the robot scans by moving



Camera bay on the electronics housing (cover in navy)

Today the claw plus jeans is heavier than the winches can lift

- Driver limit 1.06 A → running pull ≈ 18–24 N per winch (full vs empty spool)
  - about 1.1 kg total lift at the worst point in the room
- The claw is not weighed yet: CAD says ~0.76 kg printed, plus a 0.1 kg battery
- With jeans that is above the capacity today: this is the open problem
- Options: lighten the claw, a 3:1 winch reduction (~2.7× pull), or a lighter demo load
  - next step: a luggage-scale pull test on one station decides



Source: calculation sheet (Google Sheets), “Lift budget” tab. Calculated, not measured.

Key equations

$$\Sigma T_i u_{iz} = m\,(g + a), \quad 0 < T_i \le 11.9\ \text{N} \quad \text{(worst point in the room)}$$

$$m_{claw} \approx f\,\rho\,V + m_{tube} + m_{servo} = 0.55 \times 927\ \text{g} + 198\ \text{g} + 55\ \text{g} \approx 0.76\ \text{kg}$$

05

Manufacturing

Printed PETG parts, and a heat fin cut and drilled on a manual mill

Every printed fit is designed for how FDM prints, then measured

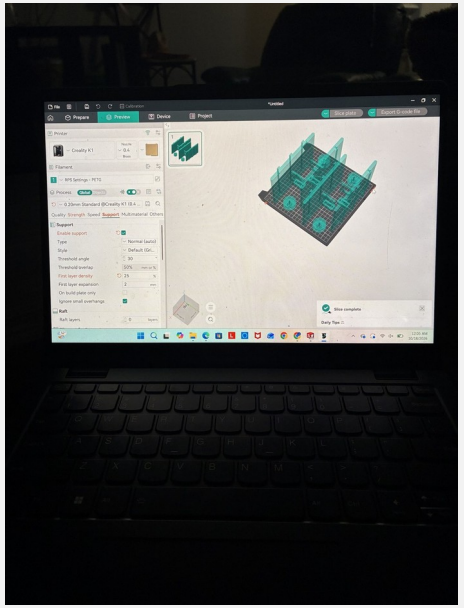
- PETG on a Creality K1 (0.4 mm nozzle, 0.2 mm layers); PLA creeps under load
- Orientation: flat face on the bed, loads along the layers, no supports on mating faces
- Printed holes come out undersize (bead rounding, shrink, first-layer squash)
  - so fits get a lead-in: 687 bearing pocket Ø14.075 with a Ø14.000 entry lip
- Heat-set M3 inserts where screws are reused
- Fewer parts: camera bay printed into the housing

Key equations

$$\text{Clearance } c = (D_{\text{hole}} - u) - D_{\text{shaft}}, \quad u = \text{printer undersize}$$

$$\text{Spool: Ø5.2 too tight} \rightarrow u \geq 0.2\ \text{mm, so Ø5.45 gives } c \approx 0.25\ \text{mm}$$

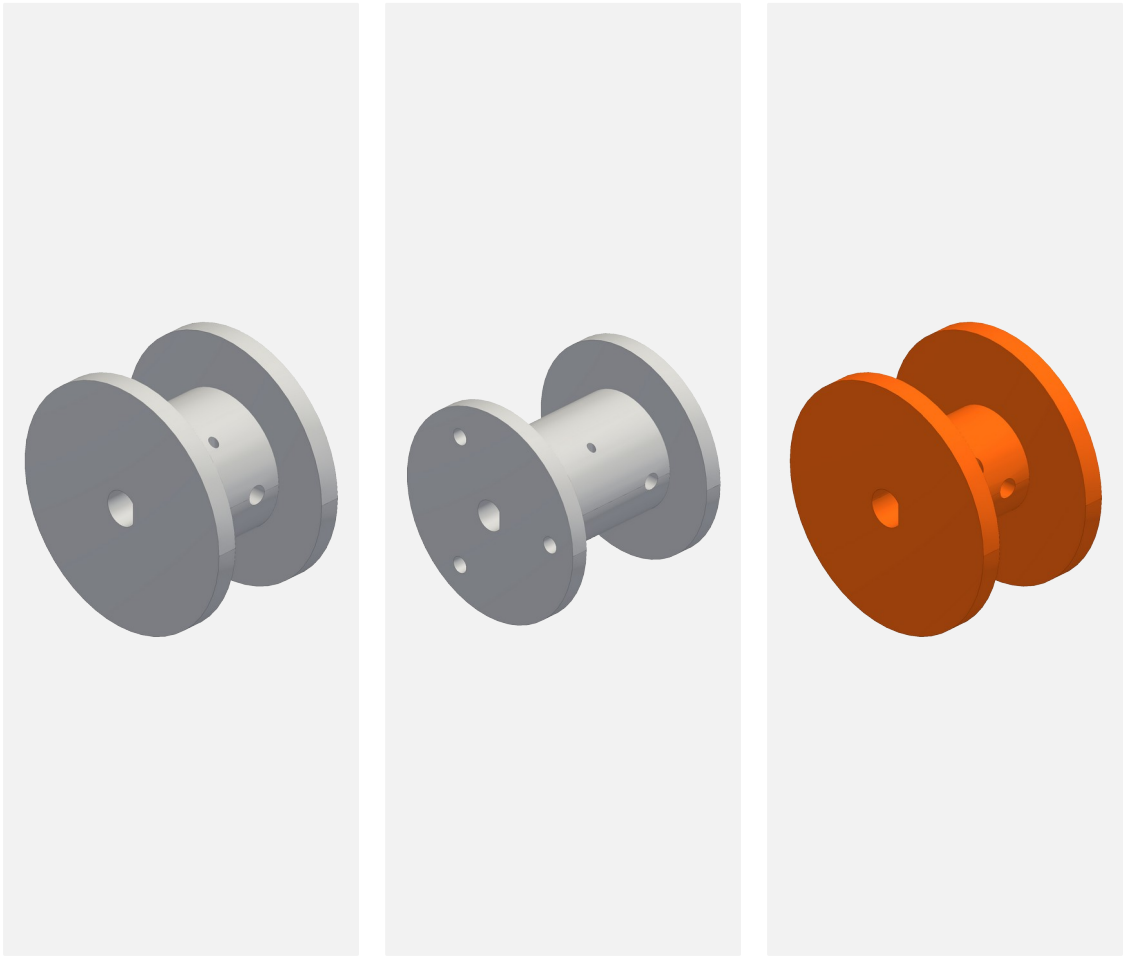




Printed PETG station shells; slicing the plate for the K1

The spool took three prints to fit the shaft and carry the torque

- v1: longer than the motor shaft could engage, so it was shortened
- v2: the Ø5.2 bore printed too tight on the 5 mm shaft
- v3: bore opened to Ø5.45 with a D-flat; the set screw on the flat carries the torque
- Ø15 core and 16 mm drum: 20 ft of line in 5 layers with 8 mm of flange margin
- Next: a fit coupon before every new interface, measured with calipers



Spool v1, v2, v3 (CAD)

The fin was sheared and drilled in one mill setup to hold ±0.1 mm

- Shop drawing first: a 1:1 US Letter print with a hole template and inch dimensions
- Sheared the blank from 0.019 in aluminium sheet: 44 × 250 mm (±0.5)
- Manual mill, one setup: all five holes located from the bottom and left edges
  - 7/8 in pilot for the motor boss and 4 × Ø3.4 on the 31 mm motor square, ±0.1 mm to each other
- Deburred, then matte black paint except the motor contact square; thermal paste there
- Test fit on the NEMA 17: boss centred, all four screws lined up









Shear and blanks, mill setup with the drawing, drilling, the finished fin

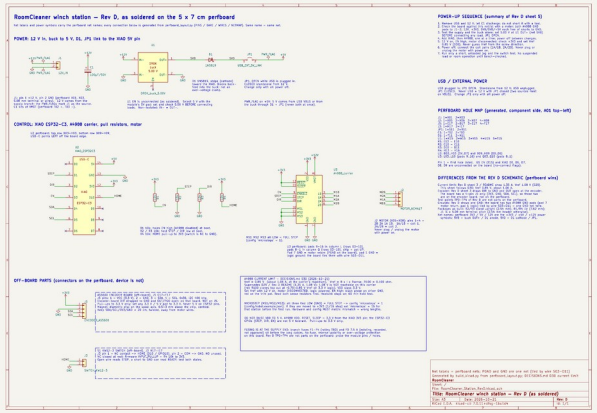
06

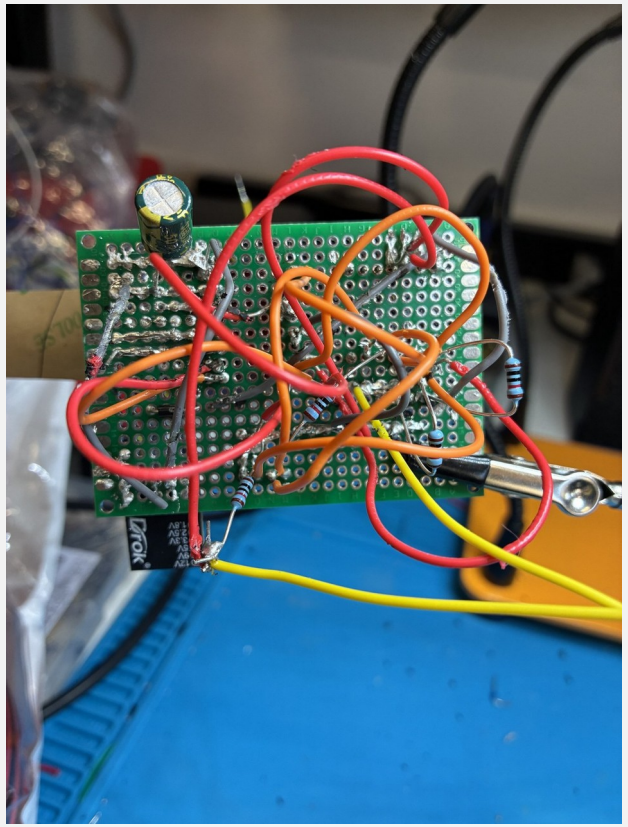
Electronics

The station board and a bring-up lesson

The station board boots safe, and its schematic matches it net for net

- The driver boots disabled (EN pull-up): a reset never steps the motor
- USB and the 5 V buck cannot back-feed each other: a link jumper plus a Schottky diode
- 100 µF at the driver's motor supply
- Connectors for the motor, homing switch, encoder and 12 V
- The KiCad schematic is generated from the soldered board's net list, so the two cannot disagree

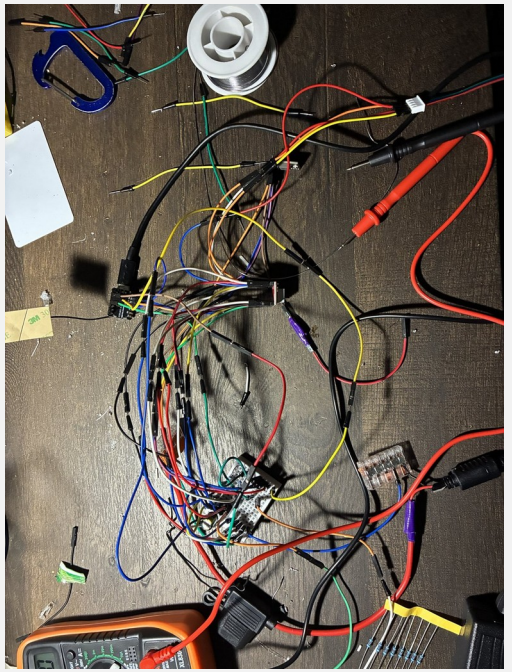




KiCad schematic (left), checked net by net against the soldered perfboard (right)

A 1.14 V logic rail, not the code, kept the first motor still

- The log said it was stepping; the shaft did not turn
- The meter said the logic rail was 1.14 V instead of 3.3 V: a bad connection
- After the fix it moved
- Since then: a meter checklist before any code
  - VDD, SLEEP, ENABLE and the buck output, every new board





Bench wiring and bring-up

07

Control logic

From the first simulator loop to the logic that runs the robot

The first logic trusted its model; the final logic measures, then checks

- Gen 1 was written against a simulator: the claw was wherever the cables were told to put it
- Real hardware breaks that: lines stretch, pulleys stick, a stepper can skip, anchors are never where the tape says
- So the final logic measures before it trusts and checks after it acts
- Each row on the right came from a problem earlier in this talk: wiring, homing, the camera
- Kept from gen 1: the cable maths, the state-machine idea, the YOLO-World detector

	First logic (gen 1)	Final logic
Where is the claw?	Assumed: where the cables were told to put it	Measured: claw camera on the floor board, refit at every startup
Room geometry	Point mass; anchors from a tape measure	Rigid claw; anchors and cable lengths fitted, with error bars
Who runs the motors	One central board steps all four	Each station runs time-stamped segments; the host coordinates
Finding laundry	Ceiling camera, pixel to floor map	Claw camera, ray from the claw's pose to the floor
Did the grab work?	Assumed once the claw closed	Camera checks pickup and delivery
Faults	E-stop only; no fault states	Supervisor: stop keeps torque, nothing restarts by itself
Where it may move	Anywhere the cables can hold	Only where the calibration earned trust

Five layers run the robot, and only the bottom one turns a motor

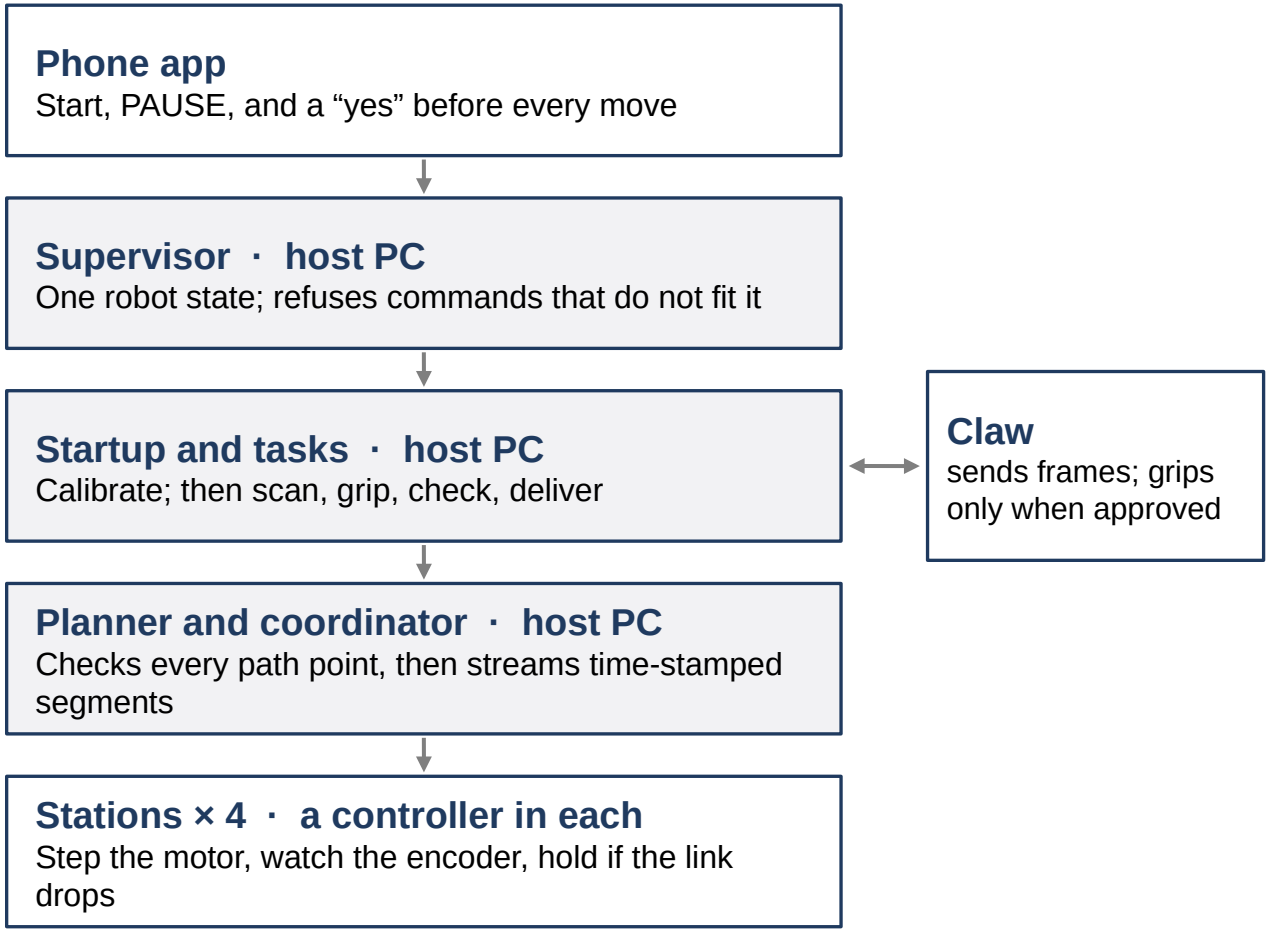
- Each layer talks only to the layers next to it
- The phone never moves a motor: it asks the supervisor
- The supervisor holds the one robot state and refuses commands that do not fit it
- Startup and tasks decide what to do; the planner checks whether it is safe to do it
- Stations run the motion on their own clocks and stop together if the host goes quiet
- The claw obeys: it grips only when the host approves

Key equations

Every path point $\mathbf{p}_k$: $L_i = \|\mathbf{a}_i - \mathbf{p}_k\|$, $T_{min} \le T_i \le T_{max}$

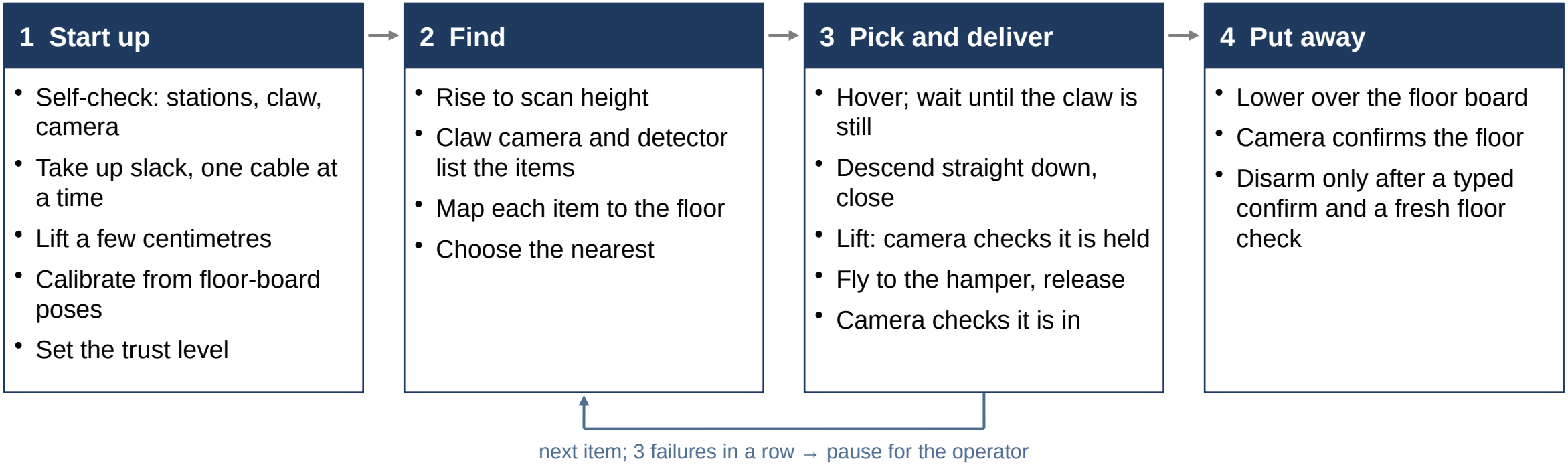
No host message for $t > t_{lease}$ → coordinated hold

Tension includes the claw, the 0.9 kg payload and 2 m/s²; lease 0.3 s in simulation



Commands flow down; status and camera frames flow back up.

One run: start up and calibrate, then pick, check and deliver each item

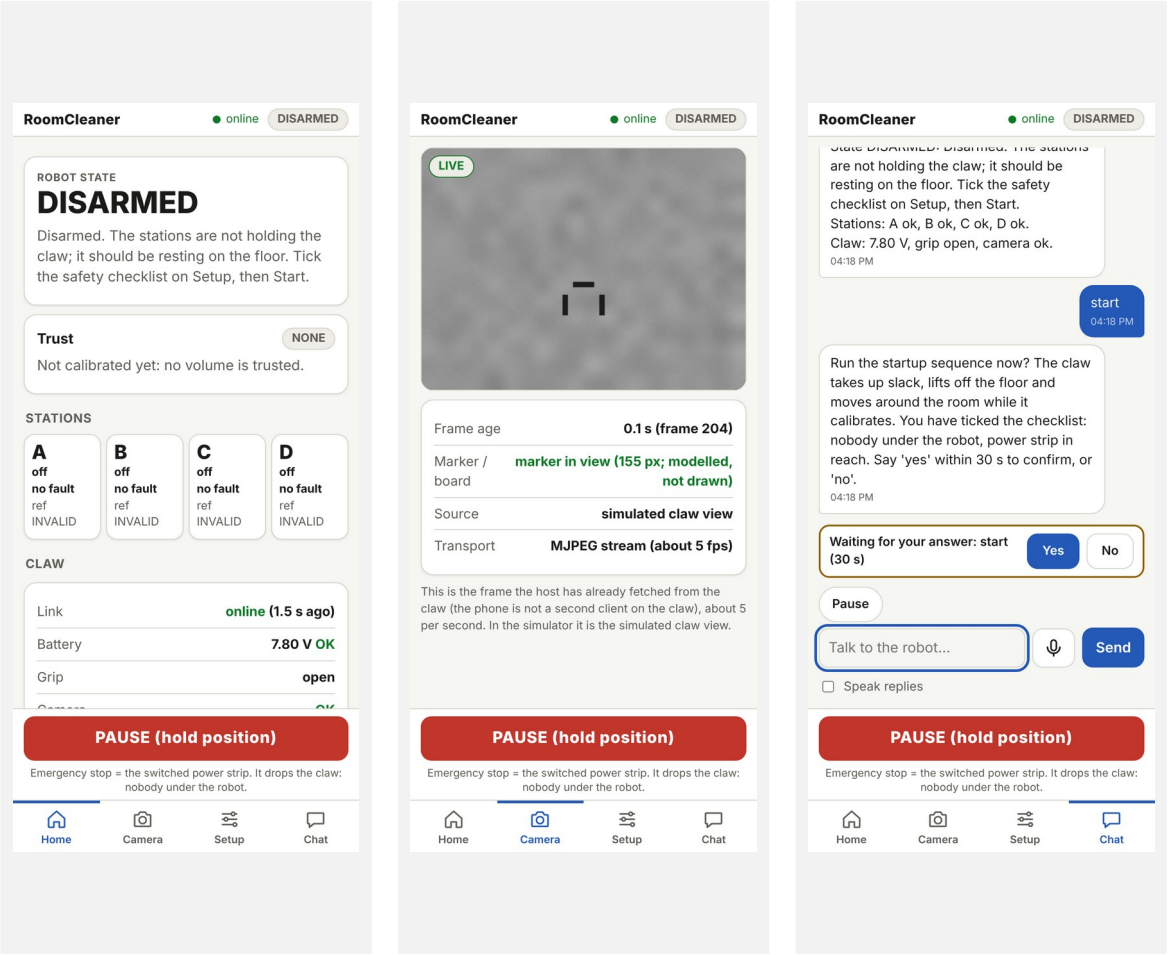


At any point: PAUSE or a fault stops every station on the same point of the path, still holding. Nothing restarts by itself; the operator lowers or recovers.

- Every step has a limit and a check; a check that fails stops the run instead of guessing
- A servo command proves nothing: the camera confirms both the pickup and the delivery

The phone is the operator's only control, and every move asks first

- Phone app over home Wi-Fi: robot status, live claw camera, guided setup, chat
- Anything that moves the robot asks “yes” first; PAUSE runs at once
- PAUSE holds position; the switched power strip is the one hard stop
- A reboot never resumes a task: the operator starts again
- Tested in a full simulator of the stations and the claw



Phone app: home, camera, chat

It plans around known obstacles but cannot yet detect new ones

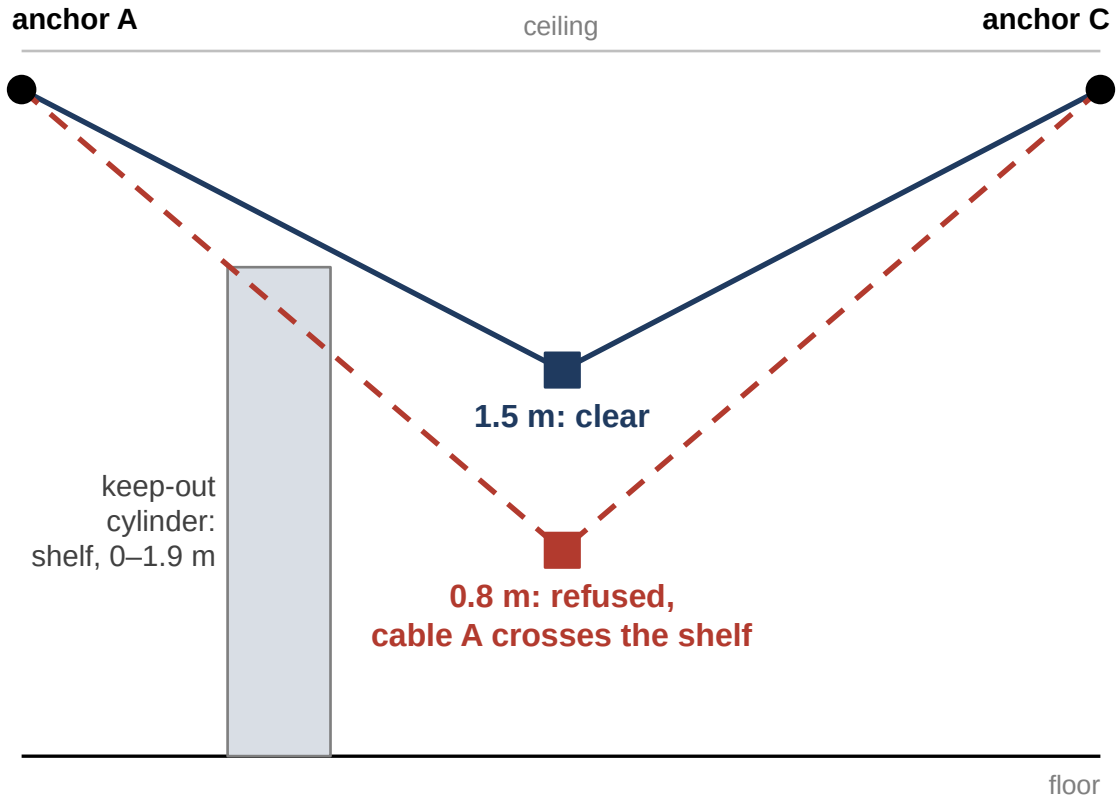
- Known obstacles (shelf, lamp, ceiling fan) are entered as keep-out cylinders
- Before any motion, every path point is checked: each cable line against each cylinder
  - plus walls 0.2 m, pulleys 0.4 m, task height 1.6 m, cable tension
- A path that fails is refused before the robot moves; it is not re-routed yet
- While moving it detects a jammed or stalled cable (encoder) and a failed grip (camera)
- It cannot yet see a person, a pet or a moved chair: the camera looks down and has no depth
  - so: nobody under the robot, PAUSE, and the power strip

Key equations

Cable i: $\mathbf{q}(s) = \mathbf{a}_i + s\,(\mathbf{b}_i - \mathbf{a}_i),\quad 0 \le s \le 1$

Hit if $z_{low} \le q_z \le z_{high}$ and $(q_x - x_c)^2 + (q_y - y_c)^2 \le r^2$

a: pulley exit; b: the cable's hole on the claw; checked at points along each cable



Side view along the A–C diagonal; same floor spot (room centre), two heights. Example obstacle, checked with the planner's own test.

08

Status and next steps

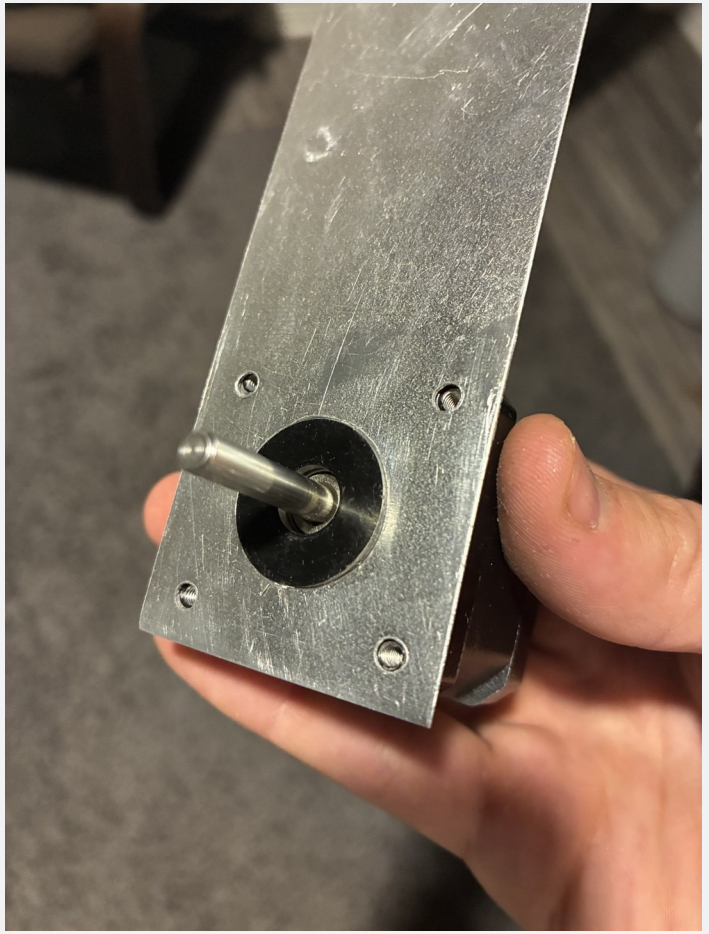
First test: self-calibration and a short move on the real hardware

- Stations up in studs, powered from the switched power strip
- Self-calibration on the real hardware: slack take-up, lift, floor-board poses, fit
- Then a short lift and a 300 mm move at 50 mm/s with the 1 lb test load
- Logged: cable lengths, encoder counts, motor temperatures and the fit's error bars
- Pass limits written before the run

Photo / video from the first test

Built and simulated; the next tests turn estimates into measurements

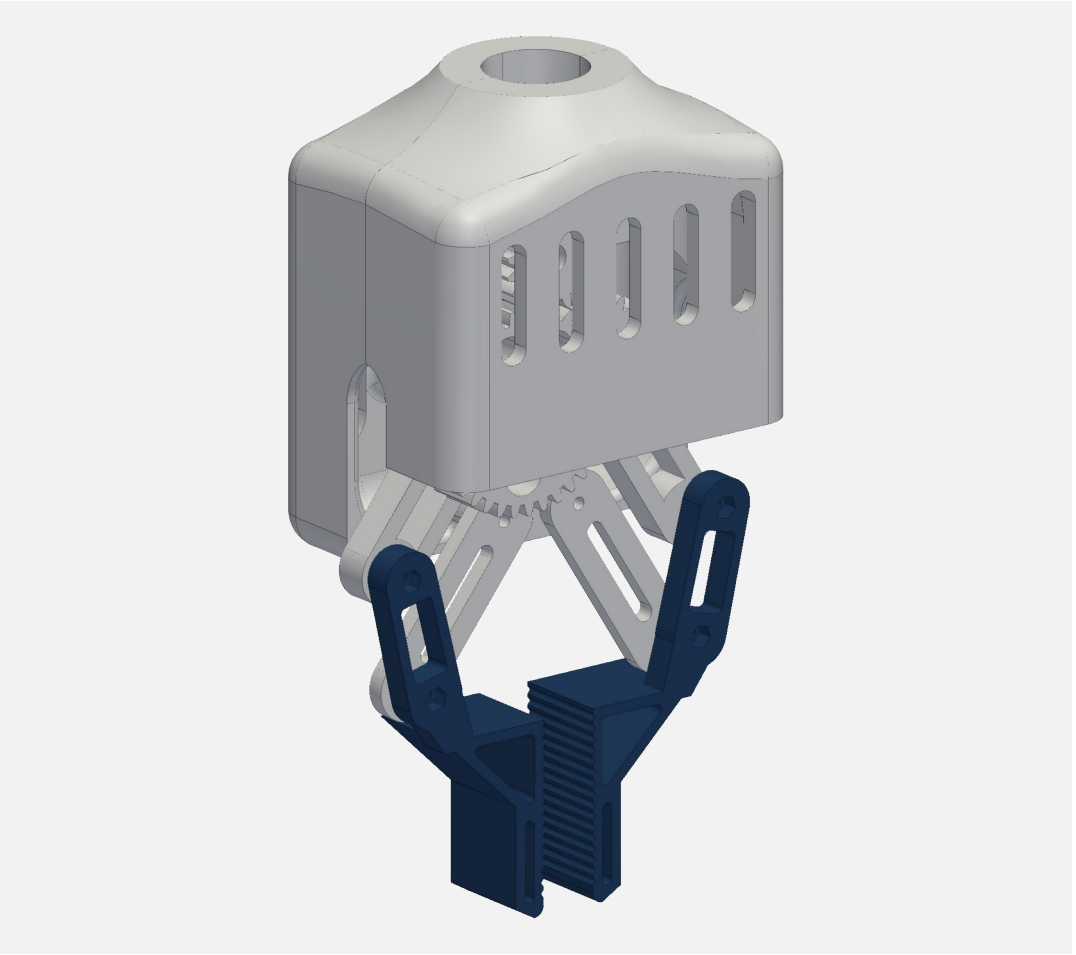
- Built: stations, station boards, heat fins, gripper parts
- Designed and checked in CAD: the hands and the camera housing
- Calibration and control: simulated, then the first test
- Next tests, with pass limits written first:
  - winch pull test (decides the capacity fix)
  - 90-min hold test: motor and printed mount under 60 °C
  - SolidWorks static and thermal studies of the station
  - four-station lift: 1 lb, 300 mm at 50 mm/s, within 2 mm per cable



Built: the heat fin test-fitted on the motor

Measure early, print test coupons first, and design for the worst case

- Measure before trusting a margin: the driver limit and the 1.14 V rail
- Print the test coupon before the part: the spool bore
- Hold tolerance where it matters: one mill setup for the fin's holes
- Design for the worst case: stall, the corner of the room, the hottest hour
- Modularity paid for itself: every station can be pulled, fixed and put back
- Question each requirement and delete parts: the IMU, the overhead camera, the five-finger hand, the long motor harnesses

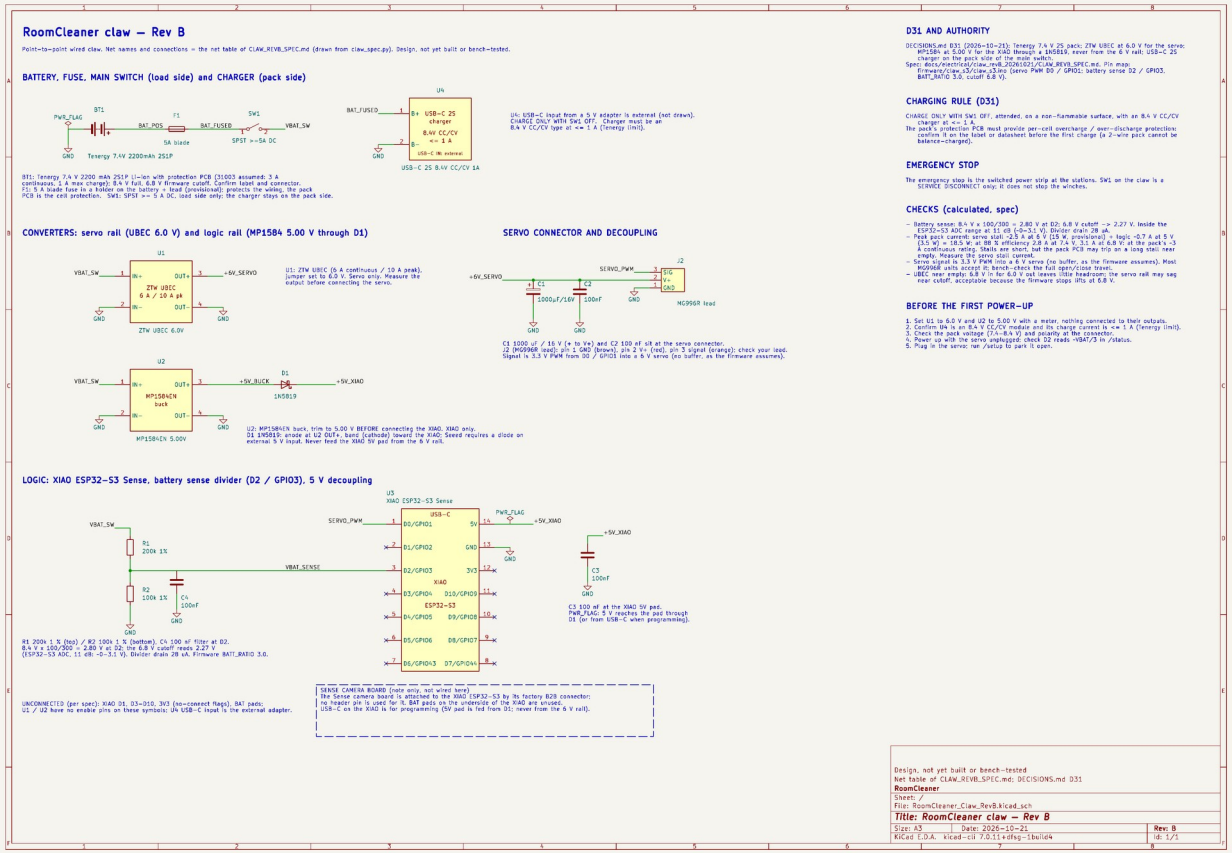


Questions

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Backup: claw electrical

- One protected 2S pack (7.4 V, 2200 mAh), 5 A fuse, main switch
- 6 V regulator for the servo only; separate 5 V buck and Schottky diode for the controller
- Battery read through 200k / 100k: 8.4 V → 2.80 V at the ADC
- Peak ≈ 2.8–3.1 A against a ≈ 3 A pack rating: stalls must stay short
- Charged on the claw only with the switch off, attended, at ≤ 1 A



Backup: gear strength (Lewis)

- Module 1.5, PETG allowable 22.5 MPa
- At 6 V servo stall: pinion SF 2.05, 36T gear 2.50, sync gear 2.00 (one jaw carrying the load)
- Module 1 was screened first: SF 0.35–0.41, so the module went up





Key equations

$$\sigma = F_t / (b\, m\, Y), \quad F_t = 2\tau / d, \quad SF = \sigma_{allow} / \sigma$$

Backup: safety

- One hard stop: the switched power strip; faults and pause hold torque
- Stations anchor into studs; the line load is mostly pull-out and prying
- Nobody stands under the robot while it moves
- Claw battery: protected pack, charged with the switch off and attended
- No safety-rated function is claimed

Backup: calculations and sources

- Every chart comes from my calculation sheet in Google Sheets, one tab per topic
  - cable tension, winch pull, heat fin, motor temperature, wiring, bead homing, floor board, lift budget, gears
- Every result is a formula on its inputs; inputs are marked, with their source or assumption
- References used:
  - Incropera, DeWitt, Bergman, Lavine: Fundamentals of Heat and Mass Transfer, ch. 3 (extended surfaces)
  - Allegro MicroSystems: A4988 datasheet (current limit)

RoomCleaner engineering calculations (Google Sheets)